Infrequent Trading, Asymmetrical Asset Price Volatility, and Welfare Loss

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Abstract

A growing share of aggregate wealth is subject to horizon-bound illiquidity through real estate holdings, private equity lock-ups, and infrequent household portfolio rebalancing. I develop a finite horizon, continuous time heterogeneous agent Lucas (1978) model to study how temporary trading restrictions affect dynamic risk sharing, asset prices, and investor welfare. Modeling an economy populated by a unit measure of risk-averse restricted investors (R) and risk-tolerant liquid traders (F), I show that a trading lock-up is nearly free when the liquid sector is well-capitalized and expensive when it is not, and that most of this cost comes from the general equilibrium price distortion the constraint creates rather than from the individual inability to trade. To self-insure against being unable to purchase equity after positive shocks during the lock-up, the restricted investor optimally hoards equity at inception. This precautionary hoarding starves the liquid trader of equity, forcing them into an under-levered position, elevating the risk-free rate, and undoing the endogenous price smoothing of the frictionless market, followed by an instantaneous block trade and a downward jump in the risk-free rate at unlock. For a benchmark four-year lock-up, when the liquid sector holds 80% of initial aggregate wealth, the lock-up is nearly costless to the restricted investor, generating a certainty equivalent welfare loss of less than 0.5%. Conversely, when the liquid sector is poorly capitalized, holding only 20% of initial wealth, the welfare loss surges more than sixfold to over 3.2%, dwarfing the classic Lucas (1987) cost of business cycle fluctuations. Decomposing this loss via a zero-measure shadow agent confirms that general equilibrium price distortions dominate the direct loss of rebalancing optionality.

1 Introduction

The foundation of continuous-time asset pricing relies on the assumption that investors can rebalance their portfolios continuously and frictionlessly. In modern financial markets, however, a substantial fraction of aggregate capital is subject to horizon-bound illiquidity. Institutional endowments commit large allocations to private equity and venture capital partnerships with strict multi-year lock-ups (Ang, Papanikolaou, and Westerfield, 2014), hedge funds suspend redemptions or erect gates during periods of market stress, corporate insiders face rigid multi-year vesting schedules, and households adjust their portfolios infrequently due to cognitive and transaction costs (Gabaix and Laibson, 2001; Abel, Eberly, and Panageas, 2007, 2013; Giglio et al., 2021), contributing to the broader macroeconomic dynamics of slow-moving capital (Mitchell, Pedersen, and Pulvino, 2007; Duffie, 2010). When a significant fraction of aggregate wealth is contractually frozen in physical shares over a finite horizon, restricted investors cannot dynamically rebalance in response to aggregate shocks, shifting the burden of market clearing entirely onto the remaining liquid traders. How do temporary trading lock-ups alter ex-ante portfolio choice and equilibrium asset prices, and what drives the resulting welfare cost?

This paper shows that a trading lock-up is nearly free when the liquid sector is well-capitalized and expensive when it is not, and that most of this welfare cost comes from the general equilibrium price distortion the constraint creates rather than from the individual inability to trade. To formalize this mechanism, I develop a finite-horizon, continuous-time heterogeneous-agent Lucas (1978) exchange economy populated by two classes of investors with Constant Relative Risk Aversion (CRRA) preferences: a more risk-averse *Restricted Investor* (R, with $\gamma_R = 4.0$) and a more risk-tolerant *Liquid Trader* or intermediary (F, with $\gamma_F = 2.0$). In the initial restricted regime ($t \in [0, T_{\text{restrict}})$), the restricted investor is locked into a fixed physical share of the Lucas tree, $\bar{\theta}_R^*$, which they choose optimally at $t = 0$ in anticipation of the constraint, while still trading the riskless bond to smooth consumption. At $t = T_{\text{restrict}}$, the lock-up expires and both agents trade frictionlessly until the terminal horizon T. Exploiting homothetic separation, I reduce the equilibrium state space to the liquid trader's wealth share $\omega_t \in (0, 1)$ and time-to-maturity $\tau = T - t$, solving the coupled nonlinear Hamilton-Jacobi-Bellman (HJB) partial differential equations via an unconditionally stable implicit upwind finite-difference scheme (Moll, 2014; Achdou et al., 2022).

The core economic mechanism operates through *ex-ante precautionary equity hoarding* and *interme-*

diary under leverage rather than a conventional leverage induced fire sale. In a frictionless market, the risk-tolerant trader F naturally levers up to hold a share of the risky asset exceeding their wealth share ($\theta_F^* > \omega_t$), borrowing from the more risk-averse investor R. Facing an upcoming lock-up, however, the restricted investor confronts an asymmetric opportunity cost: if they lock in only their modest frictionless equity share at $t = 0$, subsequent positive dividend shocks will increase their wealth while leaving their physical equity holdings frozen, diluting their exposure to the equity risk premium over the lock-up window. To self-insure against this inability to buy additional shares later, the restricted investor optimally *hoards equity ex ante* at $t = 0$. By market clearing ($\theta_F + \bar{\theta}_R^* = 1$), this hoarding starves the liquid trader F of equity, forcing F into an *under-levered* position throughout the lock-up. Because the risk-tolerant trader F strongly desires to borrow to purchase more equity against a fixed residual supply, market clearing requires an abnormally *high risk-free rate* during the restricted period to deter borrowing. At the exact moment the lock-up expires ($t = T_{\text{restrict}}$), pent-up risk-sharing demand is released: the liquid trader immediately buys a large block of equity from the restricted investor, restoring efficient risk bearing and causing the risk-free rate to *drop discontinuously*.

My analysis delivers three main quantitative and theoretical results:

First, I characterize how trading lock-ups alter return volatility and generate discontinuous gap risk at expiration. Analytically, I prove (Theorem 2) that in the frictionless baseline economy ($\gamma_R > \gamma_F > 1$), pro-cyclical wealth redistribution toward the levered, risk-tolerant trader F lowers aggregate risk aversion after positive dividend shocks; the resulting rise in the risk-free rate dampens stock price fluctuations, generating *excess price smoothness* (reducing realized stock price volatility to $\sigma_S = 8.63\%$ below the $\sigma_D = 10.0\%$ fundamental dividend volatility at $\omega_0 = 0.50$). During a trading lock-up, R's precautionary equity hoarding compresses F's leverage wedge ($\theta_F - \omega_t$), muting wealth-share volatility σ_ω and weakening this endogenous smoothing channel. Furthermore, the transition from restricted to frictionless trading produces sharp gap risk at unlock: for an equal-wealth economy ($\omega_0 = 0.50$), the expiration of the lock-up triggers an instantaneous equity block trade of roughly 13% of the aggregate market (with θ_F jumping from 63% to 76%) and an immediate downward jump in the risk-free rate of 35 to 64 basis points (exceeding 110 basis points when $\omega_0 \leq 0.10$).

Second, I show that the welfare cost of a trading lock-up to the restricted investor is highly asymmet-

ric across the wealth distribution and insensitive in its price path to restriction duration. Holding the lock-up horizon fixed at four years ($T_{\text{restrict}} = 4.0$), when the liquid sector is well-capitalized and holds 80% of initial aggregate wealth ($\omega_0 = 0.80$), the risk-tolerant trader naturally absorbs nearly the entire equity supply in both regimes, the constraint barely binds, the unlock rate jump is negligible (5 bps), and the lock-up is nearly costless, generating an ex-ante certainty equivalent welfare loss of less than 0.5%. Conversely, when the liquid sector is poorly capitalized, holding only 20% of initial wealth, precautionary hoarding distorts equilibrium prices and the restricted investor's welfare cost surges more than sixfold to 3.21%, more than thirty times the classic 0.05%–0.10% Lucas (1987) cost of business cycle fluctuations. Because i.i.d. dividend growth renders the restricted investor's optimal locked share $\bar{\theta}_R^*$ nearly horizon invariant, the magnitude of the instantaneous price distortion and unlock jump is insensitive to T_{restrict}. Longer lock-ups reduce welfare not by deepening the price distortion at any single instant, but by prolonging the duration over which agents must endure it.

Moreover, holding initial wealth equal, this vulnerability is mirrored in preference space (Figure 6): while the lock-up is completely costless along the Gorman aggregation diagonal ($\gamma_F = \gamma_R$), flipping the preference ordering so that the liquid market maker is more risk-averse than the restricted investor ($\gamma_F = 5.0, \gamma_R = 2.0$ vs. $\gamma_F = 2.0, \gamma_R = 5.0$) increases the restricted investor's welfare loss nearly eightfold from 0.39% to 3.02%.

Third, I decompose the restricted investor's ex-ante welfare loss into a *Partial Equilibrium (PE) loss of optionality*—the direct utility cost of being unable to dynamically rebalance at given prices—and a *General Equilibrium (GE) price-distortion cost*—the indirect penalty of transacting at equilibrium prices distorted by collective illiquidity. By solving an auxiliary HJB equation for a zero-measure shadow agent who shares R's preferences and trades continuously while taking the restricted economy's distorted price processes as given, I obtain an exact multiplicative decomposition of certainty-equivalent wealth retention. Across the interior of the wealth distribution, the GE price-distortion channel accounts for the majority of the total welfare loss (for example, at $\omega_0 = 0.30$, total welfare loss reaches 1.17%, of which 0.60% stems from the GE price distortion and 0.57% from the PE loss of optionality). Thus, when the liquid sector is under-capitalized, the primary danger of a lock-up is not the individual inability to trade, but the general equilibrium price distortion that the constraint inflicts on the market.

Related Literature. This paper contributes to three strands of the macro-finance and asset pricing literature. First, it builds directly on equilibrium models of limited portfolio rebalancing and slow-moving capital. Most closely related are Chien, Cole, and Lustig (2011, 2012), who study an infinite-horizon economy with active traders, intermittent rebalancers, and non-participants, demonstrating that infrequent rebalancing by passive traders amplifies the countercyclical volatility of the stochastic discount factor. While Chien, Cole, and Lustig (2011, 2012) analyze stationary dynamics using a history-dependent multiplier aggregation method, I model an explicit, finite-maturity lock-up in physical shares followed by an unlock regime switch. This finite-horizon PDE formulation allows me to solve explicitly for transition dynamics as the unlock boundary approaches, capturing the discontinuous block trade and yield curve jump at expiration and isolating the PE versus GE welfare channels. My work also complements the partial-equilibrium inattention models of Abel, Eberly, and Panageas (2007, 2013) and Ang, Papanikolaou, and Westerfield (2014) by embedding a finite non-trading window into general equilibrium, showing that endogenous price distortions outweigh the partial-equilibrium optionality loss.

It is particularly instructive to contrast my findings with Gârleanu (2009), who studies infrequent trading among CARA investors facing Brownian supply shocks and an exogenous, constant risk-free rate. In Gârleanu (2009), when constrained investors can freely choose their initial holdings entering an illiquid period, endogenous position adjustments exactly offset the illiquidity friction, leaving the equilibrium price equal to the Walrasian benchmark with no reaction at unlock dates. My model differs from Gârleanu (2009) because investors have *CRRA preferences* ($\gamma_F \neq \gamma_R$), the wealth share ω_t is an endogenous state variable, and the risk-free rate $r_f(\omega_t, t)$ clears the bond market endogenously. Consequently, even though the expected stock price path remains smooth and insensitive to lock-up duration as in Gârleanu (2009), R's precautionary equity hoarding starves F of leverage during the lock-up, elevating r_f and causing both portfolio holdings and the risk-free rate to jump discontinuously at unlock. Compared to Longstaff (2009), who derives closed form asset prices under logarithmic utility when one tree cannot be traded, my numerical framework accommodates general CRRA coefficients ($\gamma > 1$), endogenous date-zero contract selection ($\bar{\theta}_R^*$), and dynamic bond trading during the lock-up.

Second, this paper connects to the literature on heterogeneous preferences, portfolio constraints, and intermediary asset pricing (Dumas, 1989; Basak and Cuoco, 1998; Guvenen, 2009; He and Krish-

namurthy, 2013; Brunnermeier and Sannikov, 2014; Chabakauri, 2013, 2015). Whereas Chabakauri (2013, 2015) studies continuous trading subject to exogenous margin and leverage bounds, I restrict the *timing* of equity adjustments while leaving date zero share holdings unconstrained. Consistent with He and Krishnamurthy (2013), the severity of the friction scales nonlinearly with the wealth share of the unconstrained specialist (ω_0), but through an under-leverage and high-interest-rate mechanism rather than a binding margin spiral. Finally, methodologically, I extend the continuous-time finite-difference framework of Moll (2014) and Achdou et al. (2022) to a regime switching finite-horizon asset pricing environment wrapped in a global date zero contract optimization.

2 The Economy

2.1 Individuals

Consider a continuous-time finite-horizon Lucas (1978) economy with one consumption good and a unit measure of investors R (later the restricted trader) and F (later the frequent trader) with heterogeneous CRRA preferences only differ by their risk preferences and their trading technologies. The economy is defined on a filtered probability space

$$(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P}), \qquad \mathbb{F} = \{\mathcal{F}_t\}_{t\in[0,T]},$$

supporting a one-dimensional Brownian motion w_t. Individuals have standard preferences over utility flows from future consumption $c_{i,t}$ until time T discounted at some discount rate $\rho \geq 0$:

$$E_0^i\left[\int_0^T u(c_{i,t})dt + V(D, \theta_{i,T}, T)\right]$$
$$\text{where } u(c_{i,t}) = \frac{c_{i,t}^{1-\gamma_i}}{1-\gamma_i}$$

where E^i is the expectation under measure P and subject to the standard self financing budget constraint

$$dW_{it} = W_{it}(r_t + \theta_{it}(\mu_t - r_t) - c_{it})dt + W_{it}\theta_{it}\sigma_t dw_t$$
$$\frac{dW_{it}}{W_{it}} = \left[r_f + \theta_B\left(\mu_P + \frac{D_t}{P_t} - r_f\right) - \frac{C_{it}}{W_{it}}\right]dt + \theta_{it}\sigma_P dw_t \tag{1}$$

The investor trade continuously in a locally riskless bond with zero net supply and a Lucas tree (stock) with unit net supply. I will modify this assumption in the restricted trading model. The stock is a claim to some exogenous stream of dividends D_t, which follows a geometric Brownian motion (GBM) process:

$$dD_t = D_t \left(\mu_D dt + \sigma_D dw_t\right)$$

where the drift μ_D and volatility σ_D are constants.

The bond prices $B(t)$ and the stock prices $P(t)$ are determined in equilibrium.

2.2 Market Clearing

The economy is closed using a locally riskless bond in zero net supply. The Lucas tree or stock, as mentioned above, is in positive net supply and I normalize the total share of stocks at 1.

$$\theta_F + \theta_R = 1 \tag{2}$$

The total dividend at time t, D_t, is equal to the aggregate output in the economy at time t, and by market clearance, total consumption at time t.

$$c_{F,t} + c_{R,t} = D_t \tag{3}$$

The economy can be closed in many ways. I choose to have a bond in zero net supply. in equilibrium, the bonds held by the agents must equal to zero:

$$W_F(1 - \pi_i) + W_R(1 - \pi_i) = 0 \tag{4}$$

where π_i is the share of wealth held in stock.

2.3 Time Horizon

While the macro-finance literature frequently relies on infinite-horizon stationary equilibria to analyze persistent frictions, many structural constraints—such as infrequent trading and specific lock-up periods - are strictly finite in nature (Longstaff, 2009). Modeling these frictions as stationary is difficult and may not be able to generate the dynamics of agents as they approach the boundary of the

constraint. To capture this term structure of illiquidity, I solve for a time-dependent equilibrium. In my finite-horizon framework, the time-to-maturity of the friction acts as a secondary dimension of the state space, allowing us to explicitly quantify how transition dynamics dictate the evolution of risk premia and valuation ratios as the economy converges toward the unconstrained limit.

Papers like Longstaff (2009) solved GE, but relied on log utility to derive a closed form solutions for the asset prices. In this paper, I aim to use the finite-horizon continuous time framework to establish a general equilibrium solution that can be used to study the effect of constrained trading on price paths under heterogeneous risk preferences.

2.4 Terminal Values

I assume that at time T, the agent is stuck with the share of the tree, ω_T. They stop trading and simply consume their portion of the dividends forever. Their value function at T is the expected present value of this infinite consumption stream:

$$V(D, \omega, T) = \mathbb{E}_T \left[\int_T^\infty e^{-\rho(s-T)} \frac{(\omega D_s)^{1-\gamma}}{1-\gamma} ds \right]$$

Since the dividend D_s follows a Geometric Brownian Motion (GBM), its value at any future time $s > T$ is:

$$D_s = D_T \exp\left(\left(\mu_D - \frac{1}{2}\sigma_D^2 \right)(s-T) + \sigma_D (W_s - W_T) \right)$$

I substitute this into the utility function. Because ω is constant, I can pull it out of the expectation:

$$V = \frac{\omega^{1-\gamma}}{1-\gamma} \mathbb{E}_T \left[\int_T^\infty e^{-\rho(s-T)} D_s^{1-\gamma} ds \right]$$

Using the properties of the log-normal distribution, the expected value of $D_s^{1-\gamma}$ is:

$$\mathbb{E}_T[D_s^{1-\gamma}] = D_T^{1-\gamma} \exp\left((1-\gamma)\mu_D - \frac{1}{2}\gamma(1-\gamma)\sigma_D^2 \right)(s-T)$$

Now I plug this back into our integral:

$$V = \frac{\omega^{1-\gamma} D_T^{1-\gamma}}{1-\gamma} \int_T^\infty \exp\left(- \underbrace{\left[\rho - (1-\gamma)\mu_D + \frac{1}{2}\gamma(1-\gamma)\sigma_D^2 \right]}_{K} (s-T) \right) ds$$

Note that for this integral to be finite, the term in the brackets, K, must be positive. If $K > 0$, the integral of $e^{-K(s-T)}$ from T to ∞ is simply $1/K$. If $K \leq 0$, the agent is too wealthy relative to their discount rate, and the utility diverges to infinity.

$$V(D, \omega, T) = \frac{1}{K} \frac{(\omega D)^{1-\gamma}}{1-\gamma}$$

By homothetic representation (proven in the next section),

$$V(W, \omega, T) = \frac{W^{1-\gamma}}{1-\gamma} h(\omega, T), W = \omega D$$

Equating the two,

$$\frac{(\omega D)^{1-\gamma}}{1-\gamma} h(\omega, T) = \frac{1}{K} \frac{(\omega D)^{1-\gamma}}{1-\gamma} \Rightarrow h_T = \frac{1}{K}$$

But imposing the envelope condition for optimal consumption:

$$\tilde{c_T} = \frac{C_T}{W_T} = h_T^{-1/\gamma}$$

To reconcile these two:

$$h_T^{-1/\gamma} = K \Rightarrow h_T = K^{-\gamma}$$

which gives the terminal condition.

2.5 Constraints

2.5.1 No Shorting Constraint

In this economy, agents may hold any non-negative amount of stock $D_{i,t} \geq 0$ at time t, and I impose a strict no-shorting constraint on the shares of the Lucas tree for both agents.

$$\theta_i \geq 0 \quad \text{for } i \in \{F, R\}$$

Because the asset is in unit net supply ($\theta_F + \theta_R = 1$), this non-negativity constraint inherently bounds the maximum physical holdings of any individual agent to the total supply of the asset ($\theta_i \leq 1$).

Implications for the Ex-Ante Contract:

For Agent R, this no-shorting constraint strictly bounds the admissible space for the initial lock-up contract chosen at $t = 0$. Agent R cannot contract to hold a negative share, nor can they contract to hold more than 100% of the economy. Therefore, the optimization of the initial allocation is restricted to a compact admissible set:

$$\bar{\theta}_R \in [0, 1]$$

This ensures that during the restricted regime, the residual supply that Agent F must absorb, $1-\bar{\theta}_R$, is also strictly positive and less than or equal to 1.

While formally necessary to rule out arbitrage and Ponzi schemes, the Inada conditions inherent in CRRA utility ensure that this no-shorting constraint never actually binds in equilibrium. Both agents have CRRA utility, which means their marginal utility goes to infinity as wealth goes to zero ($u'(0) = \infty$). Because the Lucas tree follows a geometric Brownian motion process, its price can theoretically rise to infinity. If an agent were to short the tree ($\theta_i < 0$), there is a strictly positive probability that the asset price spikes, wiping out their wealth and pushing consumption to zero. A rational CRRA agent will never take a position that carries a risk of bankruptcy.

2.5.2 Leverage Constraints

While Agent F (the liquid market maker) is unconstrained in their ability to adjust their portfolio continuously, in reality, financial intermediaries and investors face funding liquidity limits imposed by primary brokers or regulators. I model this as a maximum leverage constraint, $\bar{m}$.

Agent F cannot borrow infinite amounts of the risk-free asset to fund their equity positions. Their portfolio weight in the risky asset is bounded by:

$$\pi_F(\omega, t) \leq \bar{m}$$

where $\bar{m} > 1$ represents the maximum allowed leverage (for example, $\bar{m} = 2$ implies a 50% margin

requirement).

When this constraint is active, Agent F's Hamilton-Jacobi-Bellman optimization becomes a constrained maximization problem. Using the Karush-Kuhn-Tucker (KKT) conditions, the optimal constrained portfolio policy takes the form of a bounded function:

$$\pi_F^{constrained} = \max\left(0, \min\left(\bar{m}, \frac{\mu_S + \delta_t - r_f}{\gamma_F \sigma_S^2} + \frac{h_{F,\omega}}{\gamma_F h_F}\frac{\sigma_\omega}{\sigma_S}\right)\right)$$

The interaction between Agent R's lock-up shares and Agent F's leverage constraint generates a non-linearity in the model. During the restricted phase, market clearing requires Agent F to hold a leverage of:

$$\pi_F(\omega, t) = \frac{1 - \bar{\theta}_R}{\omega_t}$$

As adverse macroeconomic shocks deplete Agent F's wealth ($\omega_t \to 0$), their required leverage grows unbounded. If ω_t falls to the critical threshold $\underline{\omega} = (1-\bar{\theta}_R)/\bar{m}$, the leverage constraint binds strictly.

At this boundary, the market can no longer clear under standard price dynamics. To prevent π_F from exceeding $\bar{m}$, the asset price P_t must drop. This discontinuous drop in the price level increases Agent F's wealth until the market clears exactly at the leverage constraint, serving as the driver of the tail risk in the economy.

2.6 Borrowing Constraints, Solvency, and Admissibility

Let $H_{i,t} \geq 0$ denote the time-t value of pledgeable future endowment that is not already capitalized in $W_{i,t}$. Define effective wealth by

$$X_{i,t} = W_{i,t} + H_{i,t}. \tag{5}$$

The natural debt limit is

$$W_{i,t} \geq -H_{i,t}. \tag{6}$$

If all resources are already represented by traded security wealth, as in a pure Lucas-tree exchange economy without labor income or other outside endowment, then $H_{i,t} = 0$, and effective wealth equals financial wealth. A fixed negative lower bound $W_{i,t} \geq -\overline{M}_i$ may still be imposed as an admissibility restriction, but it should not be interpreted as a natural debt limit unless a corresponding

repayment resource is specified.

Definition 2.1 (Admissible strategy)**.** Given prices, a consumption–portfolio strategy (C_i, θ_i) is admissible if:

1. C_i is nonnegative, progressively measurable, and satisfies the required utility integrability conditions;

2. φ_i is predictable and
$$\int_0^T \theta_{i,t}^2 P_t^2 \sigma_{P,t}^2 \, \mathrm{d}t < \infty \quad \text{a.s.}$$

3. wealth satisfies the self-financing equation (1);

4. effective wealth obeys $X_{i,t} \geq 0$ for every t, almost surely; and

5. for some finite constant M_i, discounted wealth plus cumulative discounted consumption is bounded below:
$$\xi_t W_{i,t} + \int_0^t \xi_s C_{i,s} \, \mathrm{d}s \geq -M_i, \qquad 0 \leq t \leq T, \tag{7}$$
where ξ is a strictly positive state-price deflator.

Condition (7) excludes doubling strategies: a strategy requiring arbitrarily large interim losses cannot remain uniformly bounded below.

2.6.1 Unrestricted equilibrium

Assumption 2.1 (Unrestricted economy)**.** The following conditions hold.

1. The horizon is finite, $D_t > 0$ almost surely, and the dividend process has sufficient positive and negative moments for all utilities and candidate state prices below to be integrable.

2. Preferences are strictly increasing and strictly concave; CRRA marginal utility satisfies
$$u_i'(c) = c^{-\gamma_i}, \qquad \lim_{c \downarrow 0} u_i'(c) = +\infty, \qquad \lim_{c \uparrow \infty} u_i'(c) = 0.$$

3. Initial effective wealth is strictly positive for both agents, and each budget set contains a strictly feasible consumption plan.

4. Admissible strategies satisfy Definition 2.1.

5. The bond and tree dynamically span the Brownian uncertainty.

6. Aggregate feasibility is

$$C_{F,t} + C_{R,t} = D_t, \qquad \theta_{F,t} + \theta_{R,t} = 1, \qquad B_{F,t} + B_{R,t} = 0. \tag{8}$$

Proposition 1 (Unrestricted equilibrium and no ruin)**.** *Under Assumption 2.1, there exists an unrestricted finite-horizon Radner equilibrium. In any equilibrium implemented from an interior Arrow-Debreu allocation,*

$$C_{i,t} > 0, \qquad X_{i,T} > 0 \quad \mathbb{P}\text{-}a.s., \tag{9}$$

and equilibrium effective wealth satisfies

$$X_{i,t} > 0, \qquad 0 \le t < T, \quad \mathbb{P}\text{-}a.s. \tag{10}$$

Consequently,

$$W_{i,t} > -H_{i,t}. \tag{11}$$

If $H_i \equiv 0$, then equilibrium financial wealth remains strictly positive. The admissibility lower bound excludes doubling strategies.

Proof. Fix strictly positive Pareto weights λ_F, λ_R. Consider the planner problem

$$\sup_{\{C_i, X_{i,T}\}} \sum_{i \in \{F,R\}} \lambda_i E\left[\int_0^T e^{-\rho_i t} u_i(C_{i,t})\, \mathrm{d}t + e^{-\rho_i T} \Phi_i(X_{i,T})\right] \tag{12}$$

subject to the pointwise resource constraints

$$C_{F,t} + C_{R,t} = D_t, \qquad X_{F,T} + X_{R,T} = \overline{X}_T, \tag{13}$$

where $\overline{X}_T > 0$ is aggregate terminal wealth. Strict concavity gives uniqueness of the allocation for fixed Pareto weights. Standard compactness or integrability arguments, together with the moment assumptions, give attainment.

Let $\xi_t > 0$ denote the multiplier on the date-t resource constraint. The first-order conditions are

$$\lambda_i e^{-\rho_i t} C_{i,t}^{-\gamma_i} = \xi_t. \tag{14}$$

Therefore,

$$C_{i,t} = \left(\frac{\lambda_i e^{-\rho_i t}}{\xi_t} \right)^{1/\gamma_i} > 0. \tag{15}$$

The terminal first-order condition similarly gives

$$\lambda_i e^{-\rho_i T} \frac{1}{K_i} X_{i,T}^{-\gamma_i} = \xi_T, \tag{16}$$

so $X_{i,T} > 0$. The Inada condition is used here to rule out a zero allocation at either margin.

The time-t value of the remaining equilibrium consumption and terminal claim is

$$X_{i,t} = \frac{1}{\xi_t} \mathbb{E}_t \left[\int_t^T \xi_s C_{i,s} \, \mathrm{d}s + \xi_T X_{i,T} \right]. \tag{17}$$

Every term on the right-hand side is nonnegative, and the terminal term is strictly positive almost surely. Since $\xi_t > 0$, equation (17) implies $X_{i,t} > 0$ for every $t < T$. By (5), this is equivalent to $W_{i,t} > -H_{i,t}$.

There is one Brownian source of uncertainty. By dynamic spanning, the bond and the tree implement every square-integrable net consumption claim generated by the Arrow-Debreu allocation. The martingale representation theorem therefore yields predictable tree holdings and residual bond holdings satisfying the self-financing budget equation. Aggregating individual implementing portfolios gives the unit tree supply and zero net bond supply because aggregate consumption equals aggregate dividends.

For an admissible strategy, the process

$$\xi_t W_{i,t} + \int_0^t \xi_s C_{i,s} \, \mathrm{d}s \tag{18}$$

is a lower-bounded local martingale and hence a supermartingale. A doubling strategy requires losses with no finite lower bound and therefore violates (7). This establishes existence, strict positivity of

effective wealth, and exclusion of doubling. □

2.6.2 Restricted Trading Constraints

In addition to the solvency constraints, for the restricted trading model, I define the Admissible Control Set $\Theta_i(t)$ as:

For Agent F (Always Liquid):

$$\Theta_F(t) = \{(c_F, \theta_F) : c_F > 0,\, 0 \leq \theta_F \leq 1\} \quad \forall t \in [0, T]$$

For Agent R (Constrained):

$$\Theta_R(t) = \begin{cases} \{(c_R, \theta_R) : c_A > 0,\, \theta_R = \theta_{R,0}\} & \text{if } 0 \leq t < T' \quad \text{(Constrained)} \\ \{(c_R, \theta_R) : c_R > 0,\, 0 \leq \theta_R \leq 1\} & \text{if } T' \leq t \leq T \quad \text{(Unconstrained)} \end{cases}$$

3 State Space

The state of the economy consists of macro variables which pin down the equilibrium price path, risk free rate, consumption path, and portfolio weights. The state variables at time t are: wealth of both types of agents, $W_{F,t}$, $W_{R,t}$, and dividend D_t. However, not all 3 state variables are needed as shown below:

3.1 Wealth Share

First, I establish that tracking (W_F, W_R) contains the exact same information as (W_{total}, ω). Let aggregate wealth in the economy be the sum of the two agents' absolute wealth:

$$W_t = W_{F,t} + W_{R,t}$$

I define the state variable, ω_t, strictly as Agent 1's share of this aggregate wealth:

$$\omega_t = \frac{W_{F,t}}{W_t}$$

By algebra, one can fully recover the absolute wealth of both agents if knowing W_t and ω_t:

$$W_{F,t} = \omega_t W_t$$
$$W_{R,t} = (1 - \omega_t) W_t$$

In a pure exchange Lucas Tree economy, the only asset available to hold is the tree itself. Therefore, the aggregate wealth of all agents must exactly equal the total market capitalization of the tree:

$$W_t = S_t$$

So, the agents' wealths are simply:

$$W_{F,t} = \omega_t S_t$$
$$W_{R,t} = (1 - \omega_t) S_t$$

3.2 Homothetic Separation

Theorem 1. *Homothetic Separation*

If the utility is CRRA and the constraints are proportional to wealth, the value function of a finite horizon Lucas Tree problem is multiplicatively separable and homogeneous of degree $1-\gamma_i$ *in absolute wealth and therefore can be written as* $V(W, \omega, t) = \frac{W^{1-\gamma}}{1-\gamma} h(\omega, t)$ *for some function* h *that does not depend on wealth* W*.*

Proof: I start by assuming the standard homothetic separation for the Value Function:

$$V(W, \omega, t) = \frac{W^{1-\gamma}}{1-\gamma} h(\omega, t)$$

Taking the partial derivatives with respect to W and ω:

$$\begin{aligned} V_W &= W^{-\gamma}h \\ V_{WW} &= -\gamma W^{-\gamma-1}h \\ V_\omega &= \frac{W^{1-\gamma}}{1-\gamma}h_\omega \\ V_{\omega\omega} &= \frac{W^{1-\gamma}}{1-\gamma}h_{\omega\omega} \\ V_{W\omega} &= W^{-\gamma}h_\omega \end{aligned}$$

Agent F's wealth evolves according to:

$$dW = W[r + \theta(\mu_S - r)]dt - Cdt + W\theta\sigma_S dw$$

The continuous-time HJB equation for Agent F, subject to a generic portfolio constraint (e.g., $\theta \geq \bar{\theta}$ or $\theta = \bar{\theta}$), is:

$$\rho V = V_t + \max_{C,\theta}\left\{\frac{C^{1-\gamma}}{1-\gamma} + V_W[W(r+\theta(\mu_S - r)) - C] + \frac{1}{2}V_{WW}W^2\theta^2\sigma_S^2 + V_\omega\mu_\omega + \frac{1}{2}V_{\omega\omega}\sigma_\omega^2 + V_{W\omega}W\theta\sigma_S\sigma_\omega\right\}$$

To solve the optimization problem for θ, I set up the Lagrangian. Let λ be the KKT multiplier associated with the portfolio constraint $g(\theta) = 0$ (or ≥ 0).

$$\mathcal{L}_\theta = V_W W\theta(\mu_S - r) + \frac{1}{2}V_{WW}W^2\theta^2\sigma_S^2 + V_{W\omega}W\theta\sigma_S\sigma_\omega + \lambda g(\theta)$$

Now, substitute the partial derivatives into the Lagrangian terms containing W:

$$\begin{aligned} V_W W &= (W^{-\gamma}h)W = W^{1-\gamma}h \\ V_{WW}W^2 &= (-\gamma W^{-\gamma-1}h)W^2 = -\gamma W^{1-\gamma}h \\ V_{W\omega}W &= (W^{-\gamma}h_\omega)W = W^{1-\gamma}h_\omega \end{aligned}$$

Notice every single term in the HJB related to the portfolio choice are scaled by exactly $W^{1-\gamma}$. Because the constraint $g(\theta)$ only restricts the weight θ and does not contain W, the only mathematical way the First Order Condition ($\frac{\partial\mathcal{L}}{\partial\theta} = 0$) can hold across all wealth levels is if the KKT multiplier λ

also scales perfectly with $W^{1-\gamma}$. Therefore, I can define a scaled, state-dependent multiplier:

$$\lambda = W^{1-\gamma}\tilde{\lambda}(\omega)$$

Substitute the value function derivatives and the scaled multiplier back into the full HJB equation:

$$\rho\frac{W^{1-\gamma}}{1-\gamma}h = \max_{\tilde{c},\theta}\left\{\frac{(\tilde{c}W)^{1-\gamma}}{1-\gamma} + W^{1-\gamma}h[r+\theta(\mu_S - r) - \tilde{c}] - \frac{1}{2}\gamma W^{1-\gamma}h\theta^2\sigma_S^2 \right.$$
$$\left. + \frac{W^{1-\gamma}}{1-\gamma}h_\omega\mu_\omega + \frac{1}{2}\frac{W^{1-\gamma}}{1-\gamma}h_{\omega\omega}\sigma_\omega^2 + W^{1-\gamma}h_\omega\theta\sigma_S\sigma_\omega + W^{1-\gamma}\tilde{\lambda}(\omega)g(\theta)\right\}$$

Because $W^{1-\gamma}$ appears in every single term on both the left and the right side of the equation, including the KKT constraint term, I can divide the entire HJB equation by $W^{1-\gamma}$:

$$\rho\frac{1}{1-\gamma}h = \max_{\tilde{c},\theta}\left\{\frac{(\tilde{c})^{1-\gamma}}{1-\gamma} + h[r+\theta(\mu_S - r) - \tilde{c}] - \frac{1}{2}\gamma h\theta^2\sigma_S^2 + \frac{1}{1-\gamma}h_\omega\mu_\omega + \frac{1}{2}\frac{1}{1-\gamma}h_{\omega\omega}\sigma_\omega^2 + h_\omega\theta\sigma_S\sigma_\omega + \tilde{\lambda}(\omega)g(\theta)\right\}$$

Therefore I can prove that $V(W,\omega,t) = \frac{W^{1-\gamma}}{1-\gamma}h(\omega,t)$ for some function h that does not depend on wealth W.

3.3 Price Dividend Ratio

Let's start from the results above:

$$V_i(W_i,\omega,D) = \frac{W_i^{1-\gamma_i}}{1-\gamma_i}h_i(\omega)$$

Because the value function is still separable, the optimal consumption-to-wealth ratio ($\tilde{c}_i$) derived from the envelope condition remains strictly a function of ω:

$$\tilde{c}_i(\omega) = h_i(\omega)^{-1/\gamma_i}$$

In equilibrium, all dividends produced by the Lucas Tree must be consumed by the two agents:

$$C_{F,t} + C_{R,t} = D_t$$

By definition, an agent's absolute consumption is their consumption rate multiplied by their absolute wealth:

$$C_{i,t} = \tilde{c}_i(\omega_t) W_{i,t}$$

Substitute this into the clearing condition:

$$\tilde{c}_F(\omega_t) W_{F,t} + \tilde{c}_R(\omega_t) W_{R,t} = D_t$$

By definition, the total wealth in the economy is simply the value of the Lucas Tree:

$$W_{F,t} + W_{R,t} = 1 \cdot P_t$$

The state variable ω_t is Agent 1's share of that total wealth:

$$W_{F,t} = \omega_t P_t, W_{R,t} = (1 - \omega_t) P_t$$

Substitute these wealth definitions back into the goods market clearing condition:

$$\tilde{c}_F(\omega_t)[\omega_t P_t] + \tilde{c}_R(\omega_t)[(1 - \omega_t) P_t] = D_t$$

Factor out the aggregate asset price P_t:

$$P_t \left[\omega_t \tilde{c}_F(\omega_t) + (1 - \omega_t) \tilde{c}_R(\omega_t)\right] = D_t$$

Because $\tilde{c}_F$ and $\tilde{c}_R$ are functions of ω_t alone, the entire bracketed term is a function of ω_t alone.

$$P_t \cdot \delta(\omega_t) = D_t$$

Divide by the dividend yield, and the proof is complete:

$$P_t = D_t \left(\frac{1}{\delta(\omega_t)} \right)$$

Let $A(\omega_t) = \frac{1}{\delta(\omega_t)}$.

$$P_t = D_t \cdot A(\omega_t)$$

By Ito's Lemma, the instantaneous return on the stock $dR_t = \frac{dP_t + D_t dt}{P_t}$ has a ex-dividend drift (μ_S) and diffusion (σ_S):

$$\sigma_S(\omega, t) = \sigma_D + \frac{A_\omega}{A}\sigma_\omega$$

$$\mu_S(\omega, t) = \mu_D + \frac{1}{A} + \frac{A_t + A_\omega \mu_\omega + \frac{1}{2}A_{\omega\omega}\sigma_\omega^2}{A} + \frac{A_\omega}{A}\sigma_D\sigma_\omega$$

For the total stock return,

$$DR_t = \frac{dP_t + D_t dt}{P_t}$$

the drift is therefore,

$$\mu_S = \mu_P + \frac{D_t}{P_t} = \mu_P + \frac{1}{A_{\omega,t}}$$

4 HJB with Two States

4.1 Complete Market HJB

Following the structure of Moll (2023), I can write the HJB equation in standard form. However, finite horizon HJB equation is also time dependent. Comparing to the infinite horizon HJB equation, $\rho V = \max\{u(c) + \mathcal{L}V\}$, I have to add the time derivative, therefore the finite horizon HJB equation has the standard form:

$$0 = \mathbf{V_t} + \max\{u(c) + \mathcal{L}V - \rho V\}$$

To solve the individual problem, I start from the standard HJB equation for 2 states:

$$\begin{aligned}\rho V_i = \max_{C_i, \theta_i} \Bigg\{ &\frac{C_i^{1-\gamma_i}}{1-\gamma_i} + V_t + V_W \left[W_i \left(r_f + \theta_i(\mu_S + \frac{D_t}{P_t} - r_f) \right) - C_i \right] + \mu_\omega V_\omega + \frac{1}{2}(W_i\theta_i\sigma_S)^2 V_{WW} \\ &+ \frac{1}{2}\sigma_\omega^2 V_{\omega\omega} + (W_i\theta_i\sigma_S\sigma_\omega)V_{W\omega} \Bigg\}\end{aligned}$$

subject to the non negativity constraint

$$\theta_i \geq 0$$

Because the utility is CRRA and the constraints are proportional, from the theorem above, the value function is multiplicatively separable. Therefore:

$$V(W_i, \omega, t) = \frac{W_i^{1-\gamma_i}}{1-\gamma_i} h_i(\omega, t)$$

I take the partial derivatives of this form directly with respect to W_i and ω:

$$\begin{aligned} V_W &= W_i^{-\gamma_i} h_i \\ V_{WW} &= -\gamma_i W_i^{-\gamma_i - 1} h_i \\ V_\omega &= \frac{W_i^{1-\gamma_i}}{1-\gamma_i} h_{i,\omega} \\ V_{W\omega} &= W_i^{-\gamma_i} h_{i,\omega} \end{aligned}$$

Taking the derivative of the HJB equation with respect to consumption C_i yields the standard Envelope condition:

$$C_i^{-\gamma_i} - V_{i,W} = 0 \implies C_i^{-\gamma_i} = W_i^{-\gamma_i} h_i$$

Defining the consumption-to-wealth ratio as $\tilde{c}_i = C_i / W_i$, I obtain the reduced FOC:

$$\tilde{c}_i = h_i^{-1/\gamma_i}$$

Next, taking the First-Order Condition with respect to the portfolio weight π_i:

$$V_{i,W} W_i (\mu_S + \delta_t - r_f) + V_{i,WW} W_i^2 \pi_i \sigma_S^2 + V_{i,W\omega} W_i \sigma_S \sigma_\omega = 0$$

Dividing by W_i and substituting the partial derivatives:

$$(W_i^{-\gamma_i} h_i)(\mu_S + \delta_t - r_f) - (\gamma_i W_i^{-\gamma_i} h_i) \pi_i \sigma_S^2 + (W_i^{-\gamma_i} h_{i,\omega}) \sigma_S \sigma_\omega = 0$$

Now, substitute the partial derivatives from the homothetic guess:

$$(W_i^{-\gamma_i} h_i)(\mu_S + \delta_t - r_f) + (-\gamma_i W_i^{-\gamma_i - 1} h_i) W_i \theta_i \sigma_S^2 + (W_i^{-\gamma_i} h_{i,\omega}) \sigma_S \sigma_\omega = 0$$

Notice that $W_i^{-\gamma_i}$ appears in every single term. I can divide it out, eliminating wealth and therefore the dividend D from the equation:

$$h_i(\mu_S + \delta_t - r_f) - \gamma_i h_i \theta_i \sigma_S^2 + h_{i,\omega}\sigma_S\sigma_\omega = 0$$

Rearranging to solve for the optimal portfolio weight θ_i^*:

$$\theta_i^* = \frac{\mu_S + \delta_t - r_f}{\gamma_i \sigma_S^2} + \frac{h_{i,\omega}}{\gamma_i h_i}\frac{\sigma_\omega}{\sigma_S}$$

Plugging back into the market clearance condition:

$$\begin{aligned}
\theta_F^* + \theta_R^* &= 1 \\
\frac{\mu_S + \delta_t - r_f}{\gamma_F \sigma_S^2} + \frac{h_{1,\omega}}{\gamma_F h_F}\frac{\sigma_\omega}{\sigma_S} + \frac{\mu_S + \delta_t - r_f}{\gamma_R \sigma_S^2} + \frac{h_{2,\omega}}{\gamma_R h_R}\frac{\sigma_\omega}{\sigma_S} &= 1 \\
\frac{r_f}{\gamma_F \sigma_S} + \frac{r_f}{\gamma_R \sigma_S} &= \frac{\mu_S + \delta_t}{\gamma_F \sigma_S} + \frac{h_{1,\omega}}{\gamma_F h_F}\sigma_\omega + \frac{\mu_S + \delta_t}{\gamma_R \sigma_S} + \frac{h_{2,\omega}}{\gamma_R h_R}\sigma_\omega \\
r_f\left(\frac{1}{\gamma_F \sigma_S} + \frac{1}{\gamma_R \sigma_S}\right) &= \frac{\mu_S + \delta_t}{\gamma_F \sigma_S} + \frac{h_{1,\omega}}{\gamma_F h_F}\sigma_\omega + \frac{\mu_S + \delta_t}{\gamma_R \sigma_S} + \frac{h_{2,\omega}}{\gamma_R h_R}\sigma_\omega
\end{aligned}$$

Now I have to solve for the price drift and volatility.

Starting with the scaled value function and the definition of the consumption-wealth ratio, I can write:

$$\begin{aligned}
\tilde{c}_F &= h_F^{-1/\gamma_F} \\
\tilde{c}_R &= h_R^{-1/\gamma_R} \\
C_F &= \tilde{c}_F W_F \\
C_R &= \tilde{c}_R W_R
\end{aligned}$$

Plugging back into the market clearance condition:

$$\begin{aligned}
\tilde{c}_F W_F + \tilde{c}_R W_R &= D \\
\theta_i^* &= \frac{\mu_S + \delta_t - r_f}{\gamma_i \sigma_S^2} + \frac{h_{i,\omega}}{\gamma_i h_i}\frac{\sigma_\omega}{\sigma_S}
\end{aligned}$$

4.2 The Utility Growth Factor

To reduce the PDE to a single spatial dimension (ω), I substitute the optimal policies ($\tilde{c}_i$ and π_i^*) and the partial derivatives back into the expanded HJB equation.

Dividing the entire equation by the scaling factor $\frac{W_i^{1-\gamma_i}}{1-\gamma_i}$ eliminates W_i, yielding:

$$\begin{aligned}\rho h_i - h_{i,t} =& \tilde{c}_i^{1-\gamma_i} + (1-\gamma_i)\left[r_f + \pi_i^*(\mu_S + \delta_t - r_f) - \tilde{c}_i\right] h_i \\ &+ \mu_\omega h_{i,\omega} - \frac{1}{2}\gamma_i(1-\gamma_i)(\pi_i^*\sigma_S)^2 h_i + \frac{1}{2}\sigma_\omega^2 h_{i,\omega\omega} + (1-\gamma_i)\pi_i^*\sigma_S\sigma_\omega h_{i,\omega}\end{aligned}$$

To organize this equation into a standard parabolic form, I collect all terms that scale linearly with the value function h_i into an endogenous scalar, which I define as the Utility Growth Factor, $G_i(\omega, t)$:

$$G_i(\omega,t) \equiv (1-\gamma_i)\left[r_f + \pi_i^*(\mu_S + \delta_t - r_f) - \tilde{c}_i\right] - \frac{1}{2}\gamma_i(1-\gamma_i)(\pi_i^*\sigma_S)^2$$

$$G_i(\omega,t) \equiv (1-\gamma_i)\Big[\underbrace{r_f + \pi_i(\mu_S - r_f)}_{\text{Expected Portfolio Return}} - \underbrace{\tilde{c}_i}_{\text{Consumption}}\Big] - \underbrace{\frac{1}{2}\gamma_i(1-\gamma_i)\pi_i^2\sigma_S^2}_{\text{Variance Penalty}}$$

G_i acts as a risk-adjusted capital accumulation rate. As an agent takes on leverage (π_i), the linear expected return increases G_i, but the quadratic variance penalty decreases G_i. The Transversality Condition requires that, in equilibrium, $G_i < \rho$; otherwise, the agent's expected lifetime utility diverges to positive infinity.

4.3 Reduced-Form HJB

Next, I take the drift of the state variable (μ_ω) and the cross-partial drift generated by the covariance between wealth and the state variable. I define this combined term as the Effective Drift, μ_ω^{eff}:

$$\mu_\omega^{\text{eff}}(\omega,t) \equiv \mu_\omega + (1-\gamma_i)\pi_i^*\sigma_S\sigma_\omega$$

Therefore, by substituting G_i and μ_ω^{eff} back into the equation, I can rewrite the complete market baseline HJB equation. Finally, to accommodate the finite-horizon structure, the PDE transitions from time t to time-to-maturity $\tau = T - t$. By the chain rule, $\frac{\partial h_i}{\partial t} = -\frac{\partial h_i}{\partial \tau}$.

The continuous-time HJB equation for the unconstrained baseline is therefore:

$$\frac{\partial h_i}{\partial \tau} + \rho h_i = \tilde{c}_i^{1-\gamma_i} + G_i(\omega, \tau) h_i + \mu_\omega^{\text{eff}}(\omega, \tau) \frac{\partial h_i}{\partial \omega} + \frac{1}{2} \sigma_\omega^2 \frac{\partial^2 h_i}{\partial \omega^2}$$

This formulation isolates the non-linear economic optimizations inside G_i and μ_ω^{eff}, allowing the PDE to be efficiently discretized and solved via the implicit upwind finite difference scheme.

4.4 Restricted Trading HJB

I now consider an economy populated by two heterogeneous agents: a Frequent trader (Agent F) and a Restricted trader (Agent R) as above but now they will face different trading restrictions. Let ω_t denote the wealth share of Agent F, such that the wealth share of Agent R is $1 - \omega_t$. The agents differ in their risk aversion ($\gamma_F \neq \gamma_R$) and in their trading technology, that is their ability to dynamically rebalance their portfolios.

The model is defined by two distinct temporal regimes:

The Restricted Regime ($0 \leq t < T_{restrict}$): Agent R is contractually locked into a fixed share of the asset, $\bar{\theta}_R$. Agent F remains fully liquid.

The Post-Restriction Regime ($t \geq T_{restrict}$): The lock-up expires, and both agents are free to trade continuously in a complete market.

4.4.1 Post Restriction Period

Let $V_i^{(2)}(W_i, \omega)$ denote the value function of agent i in the post-expiration regime ($t \geq T_{restrict}$). In this regime, both agents are unconstrained and trade to share risk. The economy is identical to the complete market case discussed above and the Hamilton-Jacobi-Bellman (HJB) equations for $i \in \{F, R\}$ are given by the standard elliptic partial differential equations as above:

$$\rho V_i^{(2)}(W_i, \omega) = \max_{c_i, \pi_i} \left\{ \frac{c_i^{1-\gamma_i}}{1-\gamma_i} + \mathcal{L}^{(2)} V_i^{(2)}(W_i, \omega) \right\}$$

where $\mathcal{L}^{(2)}$ is the infinitesimal generator under the unconstrained endogenous price processes $r_f^{(2)}(\omega)$, $\mu_S^{(2)}(\omega)$, and $\sigma_S^{(2)}(\omega)$.

$$h_F(\omega, T_{restrict}) = h_F^{(unc)}(\omega) \quad \text{and} \quad h_R(\omega, T_{restrict}) = h_R^{(unc)}(\omega)$$

4.4.2 Restricted Trading Period

During the restricted trading period, we have to write the HJB equation separately for each type of agent, we will start with agent F which the individual problems looks identical to the baseline case and therefore the same reduced form HJB:

$$\frac{\partial h_F}{\partial \tau} + \rho h_F = \tilde{c}_F^{1-\gamma_F} + G_F(\omega, \tau) h_F + \mu_\omega^{\text{eff}}(\omega, \tau) \frac{\partial h_F}{\partial \omega} + \frac{1}{2}\sigma_\omega^2 \frac{\partial^2 h_F}{\partial \omega^2}$$

However, agent F can only trade the riskless asset and consumption

$$\begin{aligned}\rho V_R = \max_{C_R} \Bigg\{ & \frac{C_i^{1-\gamma_i}}{1-\gamma_i} + V_t + V_W \left[W_i \left(r_f + \theta_i(\mu_S + \frac{D_t}{P_t} - r_f) \right) - C_i \right] + \mu_\omega V_\omega + \frac{1}{2}(W_i \theta_i \sigma_S)^2 V_{WW} \\ & + \frac{1}{2}\sigma_\omega^2 V_{\omega\omega} + (W_i \theta_i \sigma_S \sigma_\omega) V_{W\omega} \Bigg\}\end{aligned}$$

where

$$\theta_R = \bar{\theta}_R$$

Now I can take the FOC over the only choice here, consumption C_R:

$$C_R^{-\gamma_R} - V_{R,W} = 0 \implies C_R^{-\gamma_R} = W_R^{-\gamma_R} h_R$$

and following the earlier notation we can write the consumption ratio as

$$\tilde{c}_R = h_R^{-1/\gamma_R}$$

Just as what I did in the baseline case, using the homothetic separation theorem, subsititue in the value function and the partial derivatives, the HJB equation becomes:

$$\begin{aligned}\rho h_R - h_{R,t} = \tilde{c}_R^{1-\gamma_R} + (1-\gamma_R)\left[r_f + \pi_R(\omega_t)(\mu_S + \delta_t - r_f) - \tilde{c}_R\right] h_R \\ + \mu_\omega h_{R,\omega} - \frac{1}{2}\gamma_R(1-\gamma_R)(\pi_R(\omega_t)\sigma_S)^2 h_R \\ + \frac{1}{2}\sigma_\omega^2 h_{R,\omega\omega} + (1-\gamma_R)\pi_R(\omega_t)\sigma_S\sigma_\omega h_{R,\omega}\end{aligned}$$

Recall the growth factos I defined earlier, G_i, if I gather the same terms here the resulting function is

$$G_R(\omega,t) = (1-\gamma_R)\left[r_f + \left(\frac{\bar{\theta}_R}{1-\omega_t}\right)(\mu_S + \delta_t - r_f) - \tilde{c}_R\right] - \frac{1}{2}\gamma_R(1-\gamma_R)\left(\frac{\bar{\theta}_R}{1-\omega_t}\sigma_S\right)^2$$

Similarly, I can define the effective drift as before

$$mu_\omega^{\text{eff},R}(\omega,t) = \mu_\omega + (1-\gamma_R)\left(\frac{\bar{\theta}_R}{1-\omega_t}\right)\sigma_S\sigma_\omega$$

and using the same time to go notation $\tau = T - t$, the reduced form HJB equation becomes

$$\frac{\partial h_R}{\partial \tau} + \rho h_R = \tilde{c}_R^{1-\gamma_R} + G_R(\omega,\tau)h_R + \mu_\omega^{\text{eff},R}(\omega,\tau)\frac{\partial h_R}{\partial \omega} + \frac{1}{2}\sigma_\omega^2\frac{\partial^2 h_R}{\partial \omega^2}$$

which is structurally identical to the baseline HJB equation.

4.4.3 Initial Condition

At time $t = 0$, agents are endowed with shares of the Lucas tree, normalized to 1

$$e_R + e_F = 1$$

and bond endowment a_i

$$a_R + a_R = 0$$

Immediately before trading, wealth is

$$W_{i,0^-} = P_0 e_i$$

4.4.4 The Ex-Ante Optimization at Time 0

At $t = 0$, agent i chooses stock shares n_i and bond holdings B_i, subject to

$$P_0\theta_i + B_i = P_0 e_i + a_i. \tag{19}$$

The date-zero transaction is self-financing, and hence

$$W_{i0^+} = W_{i0^-}.$$

During the lock-up interval, stock holdings are fixed:

$$\theta_{R,t} = \bar{\theta_R}, \qquad 0 < t < T_L. \tag{20}$$

Bond positions and consumption may continue to adjust subject to self-financing and admissibility.

For a given candidate equilibrium price system $\mathcal{P}$, define agent i's restricted continuation value by

$$\mathcal{J}_i(n_i, b_i; \mathcal{P}) = \sup_{(C_i,B_i)\in\mathcal{A}_i^L(n_i,b_i;\mathcal{P})} \mathbb{E}\left[\int_0^{T_L} e^{-\rho_i t} u_i(C_{it})\, dt + e^{-\rho_i T_L} V_i^U\left(W_{iT_L}, D_{T_L}, T_L\right)\right] \tag{21}$$

where $\mathcal{A}_i^L$ is the set of admissible strategies satisfying (20), and V_i^U is the unrestricted continuation value at the end of the lock-up.

Agent i's date-zero problem is

$$\max_{\theta_{i,0},B_{i,0}} \mathcal{J}_i(\theta_{i,0}, B_{i,0}; \mathcal{P}) \quad \text{subject to} \quad P_0\theta_{i,0} + B_{i,0} = P_0 e_i + a_i. \tag{22}$$

For an interior solution, the first-order conditions are

$$\mathcal{J}_{i,n} = \lambda_i P_0, \qquad \mathcal{J}_{i,b} = \lambda_i.$$

Consequently,

$$\frac{\mathcal{J}_{i,n}}{\mathcal{J}_{i,b}} = P_0 \tag{23}$$

Thus, the marginal continuation value of one additional stock share, measured in units of the bond,

equals the stock price.

A date zero equilibrium satisfies

$$\theta_F + \theta_R = 1, \tag{24}$$

$$B_F + B_R = 0. \tag{25}$$

Given the individual budget constraints and aggregate endowments, (24) implies (25).

5 Numerical Algorithm

5.1 Continuous Time Heterogeneous Agent Model

Continuous time heterogeneous agent models are usually solved by finite difference methods. To solve the finite-horizon Hamilton-Jacobi-Bellman equations, I employ an implicit finite difference method following the continuous-time numerical techniques formalized by Achdou et al. (2022) and Moll (2014). Because the problem is solved backward in time from the terminal valuation at $t = T$, I define time-to-maturity as $\tau = T - t$, such that the problem is initialized at $\tau = 0$ with the terminal value function $h_i(\omega, T) = K_i^{-\gamma_i}$, and stepped forward in τ.

The reduced-form HJB equation for agent i can be written as a parabolic partial differential equation:

$$\frac{\partial h_i}{\partial \tau} + \rho h_i = u(c_i) + G_i(\omega, \tau) h_i + \mu_\omega^{\text{eff}}(\omega, \tau) \frac{\partial h_i}{\partial \omega} + \frac{1}{2} \sigma_\omega^2(\omega, \tau) \frac{\partial^2 h_i}{\partial \omega^2}$$

5.2 State Space Discretization and Upwind Scheme

I discretize the state space of the wealth distribution $\omega \in (0, 1)$ into a uniformly spaced grid with J nodes, where $\omega_j = \omega_{\min} + (j-1)\Delta\omega$ for $j = 1, \ldots, J$, and step size $\Delta\omega$. Let h_j^n denote the value of the scaled value function at state node j and time step n.

To approximate the first derivative $\partial h / \partial \omega$, we use an upwind finite difference scheme. This is critical for preserving the stability of the transition matrix. I define the forward and backward difference operators as:

$$\partial_\omega^F h_j = \frac{h_{j+1} - h_j}{\Delta\omega}, \quad \partial_\omega^B h_j = \frac{h_j - h_{j-1}}{\Delta\omega}$$

The choice of approximation depends strictly on the sign of the endogenous drift $\mu_{\omega,j}^{\text{eff}}$. I define the positive and negative components of the drift as $\mu_j^+ = \max(\mu_{\omega,j}^{\text{eff}}, 0)$ and $\mu_j^- = \min(\mu_{\omega,j}^{\text{eff}}, 0)$. The upwind approximation ensures information propagates in the correct direction across the grid:

$$\mu_\omega^{\text{eff}} \frac{\partial h_j}{\partial \omega} \approx \mu_j^+ \partial_\omega^F h_j + \mu_j^- \partial_\omega^B h_j$$

The second derivative is approximated using a standard central difference:

$$\frac{\partial^2 h_j}{\partial \omega^2} \approx \frac{h_{j+1} - 2h_j + h_{j-1}}{(\Delta\omega)^2}$$

5.3 Transition Matrix and Boundary Conditions

Substituting the upwind approximations back into the spatial terms of the HJB equation, I group the coefficients for h_{j-1}, h_j, and h_{j+1} to form the transition probabilities. The continuous Itô drift and diffusion map directly into a discrete-state Markov chain.

Let Y_j, Z_j, and X_j represent the lower-diagonal, main-diagonal, and upper-diagonal coefficients, respectively:

$$\begin{aligned} Y_j &= -\frac{\mu_j^-}{\Delta\omega} + \frac{\sigma_{\omega,j}^2}{2(\Delta\omega)^2} \\ X_j &= \frac{\mu_j^+}{\Delta\omega} + \frac{\sigma_{\omega,j}^2}{2(\Delta\omega)^2} \\ Z_j &= -\frac{\mu_j^+}{\Delta\omega} + \frac{\mu_j^-}{\Delta\omega} - \frac{\sigma_{\omega,j}^2}{(\Delta\omega)^2} \end{aligned}$$

By construction, $Z_j = -(X_j + Y_j)$. This ensures that the rows of the resulting transition matrix $\mathbf{A}$ sum to zero, a necessary property for a valid Poisson intensity matrix representing probability transitions.

To prevent the wealth distribution from crossing the boundaries $\omega = 0$ and $\omega = 1$, I impose reflecting boundary conditions, enforcing $\partial h/\partial \omega = 0$ at the extremes. In the tridiagonal matrix, we prevent outward transitions by setting $Y_1 = 0$ and $Z_1 = -X_1$ at the lower boundary ($j = 1$), and $X_J = 0$ and $Z_J = -Y_J$ at the upper boundary ($j = J$).

The full sparse tridiagonal matrix $\mathbf{A}$ can be explicitly written as:

$$\mathbf{A} = \begin{bmatrix} -X_1 & X_1 & 0 & 0 & \dots & 0 \\ Y_2 & -(Y_2 + X_2) & X_2 & 0 & \dots & 0 \\ 0 & Y_3 & -(Y_3 + X_3) & X_3 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & Y_{J-1} & -(Y_{J-1} + X_{J-1}) & X_{J-1} \\ 0 & 0 & \dots & 0 & Y_J & -Y_J \end{bmatrix}$$

Following Moll (2022), the PDE evaluated at node j can thus be written in the matrix notation as:

$$\mu_\omega^{\text{eff}} \frac{\partial h_j}{\partial \omega} + \frac{1}{2}\sigma_\omega^2 \frac{\partial^2 h_j}{\partial \omega^2} \approx [\mathbf{Ah}]_j$$

5.4 The Implicit Scheme

To solve the system, I discretize time using a backward Euler implicit scheme with step size $\Delta\tau$. The implicit formulation evaluates the non-linear utility growth factor (G_i^n) and utility flows (u_i^n) at the known time step n, but evaluates the spatial derivatives at the unknown future step $n+1$.

$$\frac{h_i^{n+1} - h_i^n}{\Delta\tau} + \rho h_i^{n+1} = u(c_i^n) + G_i^n h_i^{n+1} + \mathbf{A}^n h_i^{n+1}$$

Moving all h^{n+1} terms to the left-hand side, I obtain a linear system of equations:

$$h_i^{n+1} + \Delta\tau(\rho - G_i^n)h_i^{n+1} - \Delta\tau\mathbf{A}^n h_i^{n+1} = h_i^n + \Delta\tau u(c_i^n)$$

I can express this as a matrix inversion problem:

$$\mathbf{M}^n \mathbf{h}_i^{n+1} = \mathbf{b}^n$$

where $\mathbf{b}^n = \mathbf{h}_i^n + \Delta\tau u(c_i^n)$, and the inversion matrix $\mathbf{M}^n$ is defined as:

$$\mathbf{M}^n = \mathbf{I}\big(1 + \Delta\tau(\rho - \mathbf{G}_i^n)\big) - \Delta\tau\mathbf{A}^n$$

Because I use an upwind scheme, the off-diagonal elements of $\mathbf{M}^n$ are strictly non-positive, and as long as the transversality condition holds ($G_i < \rho$), the main diagonal is strictly positive and dominant. Therefore, $\mathbf{M}^n$ is an M-Matrix, or Poisson transition matrix. This mathematical property guarantees that the matrix inverse exists, contains only non-negative elements, and yields an unconditionally stable numerical solution free of oscillations, regardless of the chosen time step $\Delta\tau$.

5.5 Bond Position Recovery

Since I closed the general equilibrium model by using a risk-free bond in zero net supply. Every dollar borrowed by one agent must be exactly matched by a dollar lent by the other.Let B_F and B_R be the absolute dollar amount of wealth held in bonds by Agent F and Agent R. An agent's bond holding is simply their total wealth multiplied by the fraction of their portfolio not invested in the stock:

$$B_i = W_i(1 - \pi_i)$$

For Agent F, their total wealth is $W_F = \omega_t S_t$, and their leverage is $\pi_F = \frac{\theta_F}{\omega_t}$. Substituting this in:

$$B_F = \omega_t S_t \left(1 - \frac{\theta_F}{\omega_t}\right)$$

$$B_F = S_t(\omega_t - \theta_F)$$

For Agent 2, their wealth is $W_R = (1-\omega_t)S_t$, and their leverage is $\pi_R = \frac{\theta_R}{1-\omega_t}$. Substituting this in:

$$B_R = (1 - \omega_t)S_t \left(1 - \frac{\theta_R}{1 - \omega_t}\right)$$

$$B_R = S_t(1 - \omega_t - \theta_R)$$

Because the asset market clears ($\theta_F + \theta_R = 1$), Agent 2's bond holdings simplify to:

$$B_R = S_t(\theta_F - \omega_t) = -B_F$$

5.6 Welfare Calculation and the Zero-Measure Hypothetical Agent

To quantify the economic severity of the infrequent trading friction, I calculate the ex-ante welfare loss as a Certainty Equivalent (CE) wealth variation. This represents the fraction of initial wealth Agent R would be willing to surrender to avoid the lock-up constraint. The total welfare loss Δ_{total} is defined implicitly by:

$$V_R(W_R(1-\Delta_{total}),\omega_0,0;\text{Restricted}) = V_R(W_R,\omega_0,0;\text{Baseline})$$

Using the homothetic property of the value function $V = \frac{W^{1-\gamma}}{1-\gamma}h(\omega)$, I isolate Δ_{total}:

$$\Delta_{total} = 1 - \left(\frac{h_R^{\text{Restricted}}(\omega_0,0)}{h_R^{\text{Baseline}}(\omega_0,0)}\right)^{\frac{1}{1-\gamma_R}}$$

To understand this welfare cost, I decompose it into two distinct economic channels:

- The Partial Equilibrium (PE) Effect: The direct utility loss from being unable to dynamically rebalance the portfolio.
- The General Equilibrium (GE) Effect: The indirect utility loss caused by the endogenous crashing of asset prices.

To isolate these effects mathematically, I introduce a Zero-Measure Hypothetical Agent (the "fictitious agent"). This agent is subjected to the exact same economic environment as the restricted agent, but they can trade freely and possess an infinitesimal mass in the economy (zero weight). Because they have zero wealth share, their trading does not shift the market-clearing conditions, and therefore does not distort general equilibrium prices.

I solve a third, independent PDE for this fictitious agent. They optimize their scaled value function $h_{fict}(\omega,\tau)$ by trading continuously, but their utility Growth Factor (G_{fict}) and effective drift (μ_{fict}^{eff}) are evaluated using the prices generated by the restricted trading market. I will go into this fictitious agent's optimization problem later.

By comparing the value functions, I isolate the components of the welfare loss:

- Pure Cost of Illiquidity (PE): Measured by comparing h_{fict} to $h_R^{\text{Restricted}}$. This is the penalty of the friction of purely not being able to trade.

- Total Cost (GE): Measured by comparing $h_R^{\text{Restricted}}$ to h_R^{Baseline}. This represents the total loss welfare purely by the fact that the constraint forces the asset price to shift precisely when the agent is most vulnerable, i.e. during the period they cannot adjust.

5.7 Baseline Parameters

Before characterizing the equilibrium, I define the baseline parameterization used to solve the model numerically. The parameters are chosen to align with standard asset pricing literature and U.S. macroeconomic data.

I calibrate the aggregate dividend process to match the long-run moments of U.S. real consumption and dividend growth. The expected dividend growth rate is set to $\mu_D = 2\%$ per annum, and the fundamental dividend volatility is set to $\sigma_D = 10\%$. Both agents share a common subjective discount rate of $\rho = 5\%$.

To generate an intertemporal motive for risk-sharing, I introduce preference heterogeneity. Agent F (the liquid market maker) is modeled as more risk-tolerant with a relative risk aversion of $\gamma_F = 2.0$. This captures the behavior of financial intermediaries or hedge funds willing to lever up to capture risk premia. Agent R (the restricted investor) is modeled as more risk-averse, with $\gamma_R = 4.0$, representing institutional investors or household investors that seek safer returns but are frequently subjected to contractual lock-ups or trading restrictions. Later in the paper I will test a number of combinations of the risk averseness of the liquid agent and the restricted agent to see the combined effect of trading restriction and risk aversion on welfare.

I initially set the finite horizon is set to $T = 5.0$ years but will expand on that later.

Baseline Parameter Calibration

Parameter	**Description**	**Value**
γ_F	Risk Aversion: Liquid Agent (Agent F)	2.0
γ_R	Risk Aversion: Restricted Agent (Agent R)	4.0
ρ	Discount Rate	0.05
μ_D	Expected Dividend Growth	0.02
σ_D	Dividend Volatility	0.10

The continuous state space of the wealth distribution $\omega \in (0, 1)$ is discretized into a uniform grid. To ensure numerical stability and strictly adhere to the reflecting boundary conditions, the grid

boundaries are placed asymptotically close to 0 and 1.

The implicit time-stepping algorithm runs backward from the terminal condition at $\tau = 0$ to the initial inception at $\tau = T$. I employ a fine temporal discretization equivalent to $\Delta\tau = 10^{-4}$ to ensure a smooth transition of the non-linear pricing coefficients. The algorithm loops until the endogenous asset prices and value functions converge at each time step within a some numerical tolerance of 10^{-6}.

6 Characterizing the Equilibrium

6.1 Definition of the Equilibrium

The state space of the economy operates on two levels: individual and aggregate. While an individual agent's utility and consumption level depend on their wealth $W_{i,t}$, the assumption of CRRA utility and strictly proportional constraints ensures that the equilibrium exhibits scale independence as stated above. Therefore, all equilibrium price processes, optimal portfolio weights, and consumption-to-wealth ratios are completely characterized by the reduced aggregate state space: the wealth distribution $\omega_t \in (0, 1)$ and explicit time $t \in [0, T]$.

A Recursive Competitive Equilibrium in this finite-horizon economy consists of:

- A set of strictly positive value functions $\mathcal{V} = \{V_F(W_F, \omega, t), V_R(W_R, \omega, t)\}$;
- A set of asset pricing functions $\mathcal{P} = \{r_f(\omega, t), \mu_S(\omega, t), \sigma_S(\omega, t)\}$ for the risk-free rate, expected stock return, and stock volatility, respectively;
- A set of optimal policy functions for absolute consumption and portfolio choice

  $\mathcal{A} = \{C_F(W_F, \omega, t), C_R(W_R, \omega, t), \pi_F(\omega, t)\}$;
- An initial portfolio allocation $\bar{\theta}^*_R \in (0, 1)$ chosen at $t = 0$;

such that, given the initial state $(\omega_0, W_{R,0}, t = 0)$, the following conditions hold for all $t \in [0, T)$ and $\omega_t \in (0, 1)$:

1. Individual Optimality:

   Taking the endogenous price processes $\mathcal{P}$ as given, the policy functions $\mathcal{A}$ maximize the agents' expected lifetime utility.

Agent F optimizes over both consumption and portfolio choice, such that V_F solves their unconstrained Hamilton-Jacobi-Bellman (HJB) equation.

Agent R optimizes only over consumption during the restricted trading period, while their portfolio is constrained to $\pi_R(\omega, t) = \bar{\theta}_R^*/(1-\omega_t)$, such that V_R solves their restricted HJB equation. After the trading restriction expiration, agent R also optimizes over both consumption and portfolio choice, such that V_F solves their unconstrained Hamilton-Jacobi-Bellman (HJB) equation.

At expiration $t = T$, both value functions satisfy the terminal value condition: $V_i(W_i, \omega, T) = K_i^{-\gamma_i} \frac{W_i^{1-\gamma_i}}{1-\gamma_i}$.

2. Asset Market Clearing:

   The risk-free bond is in zero net supply, and the Lucas tree is in unit net supply. At every instant, the sum of the agents' physical holdings of the tree must equal exactly one. Expressed in terms of portfolio weights as a fraction of aggregate wealth, the equity market clears:

$$\omega_t \pi_F(\omega, t) + (1-\omega_t)\pi_R(\omega, t) = 1$$

   Because Agent R's leverage is fixed by the initial contract, this condition requires Agent F to absorb the residual supply, resulting in $\pi_F(\omega, t) = (1-\bar{\theta}_R^*)/\omega_t$.

3. Goods Market Clearing:

   At every instant, aggregate consumption must equal the aggregate dividend produced by the Lucas tree:

$$C_F(\omega, t) + C_R(\omega, t) = D_t$$

   By Walras' Law, if the goods market and the equity market clear, the risk-free bond market clears automatically.

4. Consistency of State Evolution:

   The wealth distribution ω_t evolves according to Itô's lemma, with its endogenous drift (μ_ω) and diffusion (σ_ω) generated consistently by the agents' optimal policies $\mathcal{A}$ and the market-clearing price processes $\mathcal{P}$.

5. Ex-Ante Optimality:

   The initial stock lock-up $\bar{\theta}_R^*$ is chosen rationally by Agent R at time $t = 0$ to maximize their initial expected lifetime utility, anticipating the subsequent equilibrium path of prices and state transitions:

$$\bar{\theta}_R^* = \arg\max_{\bar{\theta}_R} V_R\left(W_{R,0}, \omega_0, 0; \bar{\theta}_R\right)$$

Having established the numerical solution method to the finite-horizon HJB equations, I now characterize the equilibrium price dynamics, trading behaviors, and welfare costs generated by the infrequent trading friction. Because the equilibrium is determined by a non-linear mapping from the state variables (ω, τ, W) to asset prices, I rely on the numerical solution to explore the relationship. I first hold time constant to examine the state-space dynamics, then analyze the temporal approach to the terminal state, and finally quantify the welfare costs of the trading friction.

6.2 Baseline Model

I first fix the initial wealth distribution of wealth share to 50-50 and compute the baseline model where both types of agents can trade freely across the entire time horizon. I first look at the properties of the resulting policy function before diving into the simulations.

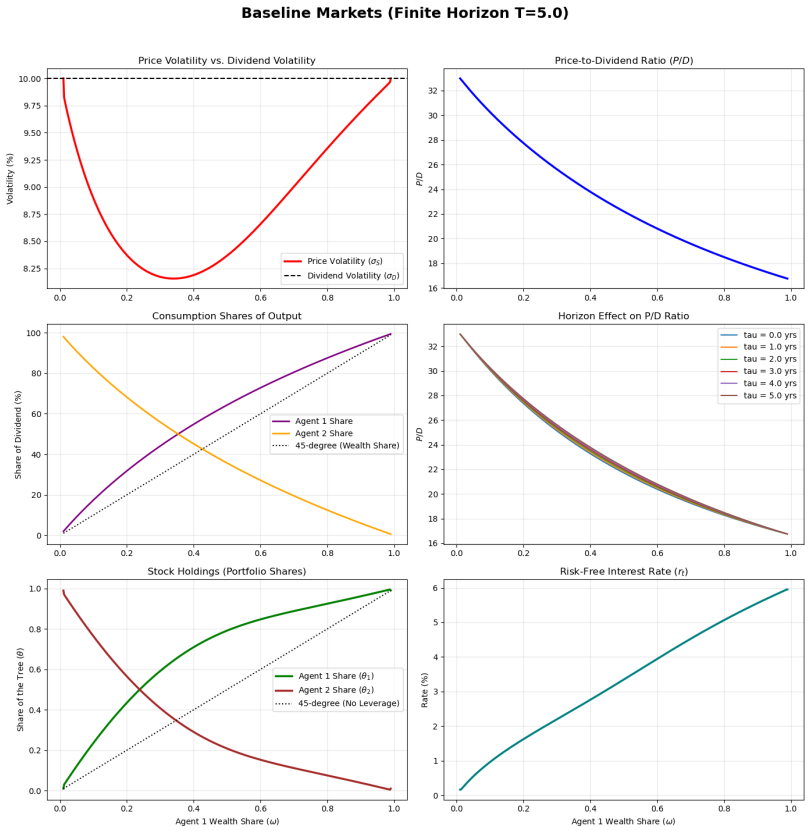


Figure 1: Baseline Model Solution and Policy Functions

Let's first look at the Stock Holdings graph. Because Agent F is more risk-tolerant ($\gamma_F < \gamma_R$), they actively lever up to capture the equity premium, as indicated by their stock holding position greater than 50% of the total share (above the 45 degree line). Agent F's physical share of the tree (θ_F) is strictly greater than their wealth share (ω) across the interior, and greater than the 45 degree line. Conversely, Agent R holds a safe portfolio, lending to Agent F through the risk-free bond market.

Next, the Risk-Free Rate and P/D ratio graphs show us the dynamics of the policy function of asset prices. As a sequence of positive shocks hit the economy and the risk-tolerant Agents F's wealth increase ($\omega \to 1$), the wealth-weighted aggregate risk aversion of the economy falls. The demand for precautionary savings decreases, driving the risk-free rate up monotonically. At the same time, the aggregate willingness to pay for the asset's cash flows drops, lowering the P/D ratio.

Then if we look at the price volatility, it highlights a feature of the complete market with CRRA coefficients $\gamma > 1$, that asset prices are less volatile than fundamentals ($\sigma_S < \sigma_D$ in the interior).

While heterogeneous agent models often generate excess volatility when markets are incomplete or feature trading frictions, the frictionless baseline generates excess smoothness (Dumas 1989). Mathematically, this is driven by the strong endogenous, pro-cyclical response of the risk-free rate. A positive dividend shock redistributes wealth to the levered, less risk-averse Agent F, lowering the economy's aggregate risk aversion. The resulting drop in precautionary savings demand causes the risk-free rate to rise sharply. This higher discount rate partially offsets the positive cash flow shock, structurally dampening the asset price response and leading to $\sigma_S < \sigma_D$.

Why does the finite horizon not destroy the standard infinite-horizon results? If we look at the horizon effect graph, the P/D ratio curves for varying times-to-maturity (τ) are tightly clustered and nearly indistinguishable. This occurs because, in a frictionless market, agents can trade continuously right up to the terminal boundary at $t = T$. They use this perfect liquidity to dynamically shift their physical holdings into the optimal buy-and-hold portfolio immediately prior to market closure. Because they transition into the terminal consition perfectly, the shadow cost of losing future trading opportunities is quantitatively small. Therefore, the finite-horizon complete market preserves the qualitative features of an infinite-horizon economy, while being mathematically and computationally more trackable.

After exploring the dynamics of the policy functions, we can now look at the results from simulating the economy over a large number of paths. I simulate Monte Carlo paths of the economy over the finite horizon initialized at $\omega_0 = 0.5$.

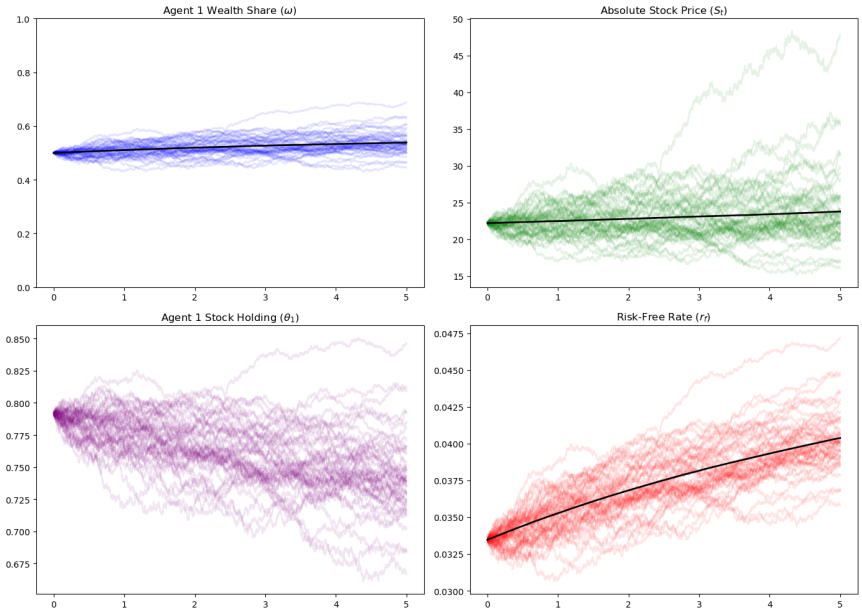


Figure 2: Baseline Monte Carlo Time-Series Simulations

From the simulation, if we look at the wealth share (ω) and risk-free rate (r_f), while individual paths are highly various, the expected trajectories suggest that there is a structural upward drift. Because Agent F bears a disproportionate share of the fundamental risk by holding a larger share of the Lucas tree, they earn the equity risk premium on average. Consequently, their expected wealth share drifts upward over time, reaching approximately 0.55 at the end of the finite horizon compared to the 0.5 at the start of time. As Agent F gradually taking on a larger share of the economy, aggregate risk aversion steadily declines, driving the expected risk-free rate upwards from 3.3% to 4.0%. The result is also consistent with the findings of Basak and Cuoco(1998), that the risk-free rate is increasing in the unrestricted investor's wealth or consumption share.

The Monte Carlo simulations also numerically validate the excess smoothness theorem derived analytically in the next section. Across the simulated sample paths, the average realized volatility of the stock price is $\sigma_S^{realized} = 8.63\%$. This is actually lower than the fundamental dividend volatility of $\sigma_D = 10\%$. This confirms that frictionless dynamic trading successfully suppresses aggregate pricing fluctuations, which I will formally proof in the section below.

To maintain the risk-sharing equilibrium, Agent F's stock holdings must fluctuate, ranging anywhere from 65% to 85% of the total asset supply depending on the sequence of realized macroeconomic

shocks, as shown in the stock holding (θ_F) panel of the graph. This visualizes the large volume of continuous rebalancing required by a frictionless market. It is precisely this continuous adjustment mechanism that is eliminated when trading restrictions are introduced, therefore motivate us to investigate the infrequent trading model further.

6.3 Excess Volatility and Smoothness in the Heterogeneous Agent Baseline Model

We allow for a general homogeneous terminal continuation value. This class includes terminal liquidation, CRRA bequests, and an infinite post-closure dividend-consumption continuation. Homogeneity preserves the one-dimensional representation ,t). Excess smoothing follows if the terminal equilibrium price-dividend boundary is nonincreasing in the risk-tolerant agent's wealth share and the normalized effective discount rate is nondecreasing in that state.

Theorem 2 (Distributional amplification and smoothing)**.** *Consider a finite-trading-horizon Lucas-tree economy on $[0,T]$ with one tree in unit supply and one bond in zero net supply. Let aggregate dividends satisfy*

$$\frac{dD_t}{D_t} = \mu_D dt + \sigma_D dZ_t, \qquad \sigma_D > 0.$$

Let $\omega_t = W_{F,t}/P_t \in (0,1)$ denote agent F's share of aggregate financial wealth, where agent F is the more risk-tolerant agent. Let $\vartheta_F(\omega,t)$ denote agent F's equilibrium ownership share of the unit-supply tree. Suppose the equilibrium tree price is homogeneous of degree one in dividends:

$$P_t = D_t A(\omega_t, t),$$

where $A(\omega,t) > 0$ is continuously differentiable in ω.

Suppose further that

$$1 - \frac{A_\omega(\omega,t)}{A(\omega,t)}\left[\vartheta_F(\omega,t) - \omega\right] > 0. \tag{26}$$

Then the signed instantaneous diffusion of the tree price is

$$\sigma_P(\omega,t) = \frac{\sigma_D}{1 - \dfrac{A_\omega(\omega,t)}{A(\omega,t)}\left[\vartheta_F(\omega,t) - \omega\right]}. \tag{27}$$

In particular, suppose that agent F is levered in the tree in the sense that

$$\vartheta_F(\omega, t) > \omega. \tag{28}$$

Then the following statements hold.

1. *If*

$$A_\omega(\omega, t) < 0,$$

then

$$0 < \sigma_P(\omega, t) < \sigma_D.$$

Thus, endogenous wealth redistribution generates excess price smoothing.

2. *If*

$$0 < \frac{A_\omega(\omega, t)}{A(\omega, t)} \left[\vartheta_F(\omega, t) - \omega\right] < 1,$$

then

$$\sigma_P(\omega, t) > \sigma_D.$$

Thus, endogenous wealth redistribution generates excess price volatility.

Proof. Since the bond is in zero net supply and the tree is in unit supply, aggregate financial wealth equals the ex-dividend value of the tree:

$$W_{F,t} + W_{R,t} = P_t.$$

Agent F's financial wealth can therefore be written as

$$W_{F,t} = \omega_t P_t.$$

Let the tree price satisfy

$$\frac{dP_t}{P_t} = \mu_P(\omega_t, t)dt + \sigma_P(\omega_t, t)dZ_t.$$

Because the bond has no instantaneous diffusion exposure and agent F owns ϑ_F shares of the

unit-supply tree, the diffusion coefficient of agent F's wealth is

$$dW_{F,t} = \cdots + \vartheta_F(\omega_t, t) P_t \sigma_P(\omega_t, t) dZ_t.$$

Applying Itô's quotient rule to

$$\omega_t = \frac{W_{F,t}}{P_t}$$

implies that the diffusion coefficient of the wealth-share process is

$$\sigma_\omega(\omega, t) = \left[\vartheta_F(\omega, t) - \omega\right] \sigma_P(\omega, t). \tag{29}$$

Next, the homogeneous price representation is

$$P_t = D_t A(\omega_t, t).$$

Applying Itô's lemma, the diffusion coefficient of the tree price is

$$\sigma_P(\omega, t) = \sigma_D + \frac{A_\omega(\omega, t)}{A(\omega, t)} \sigma_\omega(\omega, t). \tag{30}$$

Substituting equation ((29)) into equation ((30)) yields

$$\sigma_P = \sigma_D + \frac{A_\omega}{A} \left[\vartheta_F - \omega\right] \sigma_P.$$

Rearranging gives

$$\sigma_P \left\{ 1 - \frac{A_\omega}{A} \left[\vartheta_F - \omega\right] \right\} = \sigma_D.$$

Under condition ((26)), division by the term in braces is valid, and hence

$$\sigma_P = \frac{\sigma_D}{1 - \dfrac{A_\omega}{A} \left[\vartheta_F - \omega\right]}.$$

Suppose now that $\vartheta_F - \omega > 0$. If $A_\omega < 0$, then

$$-\frac{A_\omega}{A} \left[\vartheta_F - \omega\right] > 0,$$

so the denominator in equation ((27)) exceeds one. Therefore,

$$0 < \sigma_P < \sigma_D.$$

Alternatively, if

$$0 < \frac{A_\omega}{A} [\vartheta_F - \omega] < 1,$$

then the denominator in equation ((27)) lies strictly between zero and one. Therefore,

$$\sigma_P > \sigma_D.$$

This proves both claims.

The Gorman Aggregation Limit: If $\gamma_1 = \gamma_2 = \gamma$, the agents hold identical, unlevered market portfolios ($\pi_1 = \pi_2 = 1$). Consequently, the wealth distribution cannot drift or diffuse ($\sigma_\omega = 0$). Substituting this into the volatility decomposition ((30)) yields $\sigma_S = \sigma_D + 0$, proving that excess smoothness vanishes and prices perfectly track fundamentals when aggregation holds. □

Corollary 1 (Expected realized excess smoothing)**.** *Suppose the conditions of the theorem hold for all states reached on* $[t, t+H] \subset [0, T]$*. Then*

$$\frac{1}{H} \int_t^{t+H} \sigma_{S,s}^2 ds < \sigma_D^2 \quad a.s.$$

and therefore

$$\mathbb{E}_t \left[\frac{1}{H} \int_t^{t+H} \sigma_{S,s}^2 ds \right] < \sigma_D^2.$$

Corollary 2 (Monotonicity of the price-dividend ratio)**.** *Suppose CRRA preferences and homogeneous terminal continuation values imply*

$$S(t, D, \omega) = DF(t, \omega).$$

Suppose that F *solves*

$$F_t + \bar{b}(t,\omega)F_\omega + \frac{1}{2}\eta(t,\omega)^2 F_{\omega\omega} - q(t,\omega)F + 1 = 0, \qquad F(T,\omega) = f_T(\omega).$$

Assume:

1. $f_T'(\omega) \leq 0$*;*
2. $q_\omega(t,\omega) \geq 0$ *for all* (t,ω)*;*
3. *the one-dimensional diffusion generated by*

$$\bar{b}(t,\omega)\partial_\omega + \frac{1}{2}\eta(t,\omega)^2\partial_{\omega\omega}$$

preserves order.

Then

$$F_\omega(t,\omega) \leq 0.$$

The inequality is strict at interior states if either $f_T' < 0$ *on a reachable set or* $q_\omega > 0$ *on a reachable set of positive measure.*

6.4 Baseline Infrequent Trading Model

To study the effect of the trading restriction, I simulate an economy where Agent R is strictly locked into their initial allocation for the first 2.5 years of the horizon.

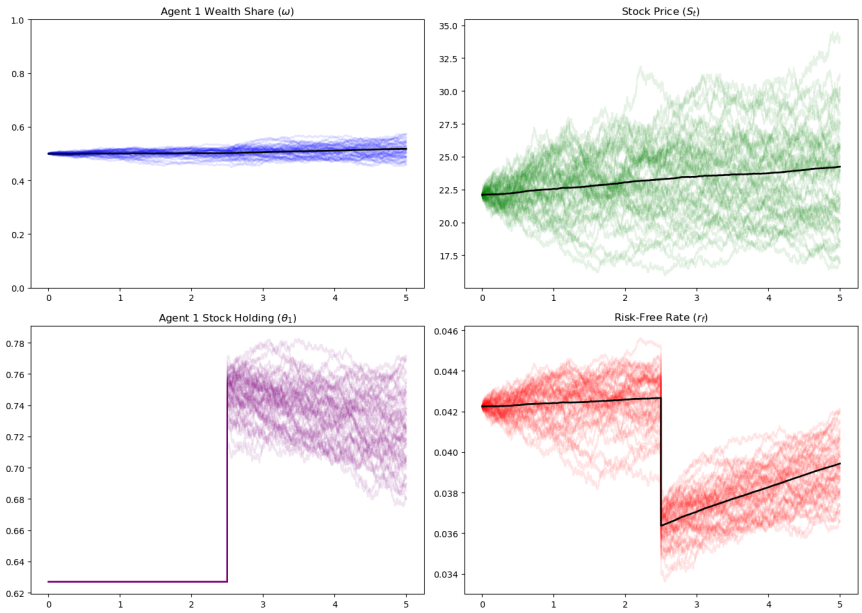


Figure 3: Baseline Infrequent Trading Monte Carlo Time-Series Simulations

During the restricted regime ($t < 2.5$), the stock holding graph shows the stock held by the agent F is completely fixed at roughly 63%. Despite the economy experiencing continuous dividend shocks as evidenced by the stock price paths, the agents are required to hold a constant share of the tree, therefore unable to dynamically $\pm 15\%$ rebalancing their portfolio as shown in the frictionless baseline model.

If we look at the moment the trading restriction expires at $t = 2.5$, there is a noticeable jump in the portfolio holdings of both type of agents. Because at the time we transition from restricted trading to free trading, the allocation (around 63%) was sub-optimal compared to Agent F's frictionless target at that wealth level, the sudden return of liquidity triggers a massive instantaneous block trade. Agent F shifts their holding upward to approximately 76%, instantly absorbing a large amount of the risky asset from Agent R.

The instantaneous block trade at $t = 2.5$ transfers a large amount of fundamental risk from the more risk-averse Agent R back to the more risk-tolerant Agent F. This sudden reallocation of risk changes the aggregate risk aversion of the economy. The demand for precautionary savings decreases, causing the risk-free rate to jump down from 4.3% down to 3.6% as seen in the riskfree rate chart. This visualizes a theoretical result that infrequent trading not only distorts prices while the constraint is active, but generates significant jump risk in asset prices the moment liquidity is restored.

6.5 The Effect of Initial Wealth Distribution

A central feature of the heterogeneous agent asset pricing literature is that financial frictions do not generate uniform economic distortions, rather, the severity depends on the wealth share of the marginal pricer (He and Krishnamurthy 2013). In our model, the magnitude of the price distortion and the magnitude of the subsequent price correction when liquidity is restored, depends endogenously on the initial wealth distribution ω_0, which represents the wealth share of the liquid agent F.

To numerically analyze this relationship, I simulate the regime-switching economy across a spectrum of initial wealth distributions. For each ω_0, the time 0 market clearance determines the lock-up contract for each agent. Figure 4 highlights two representative cases from this state space: a poor liquid agent F with a rich constrained agent R, and a rich liquid agent F with a poor constrained agent R.

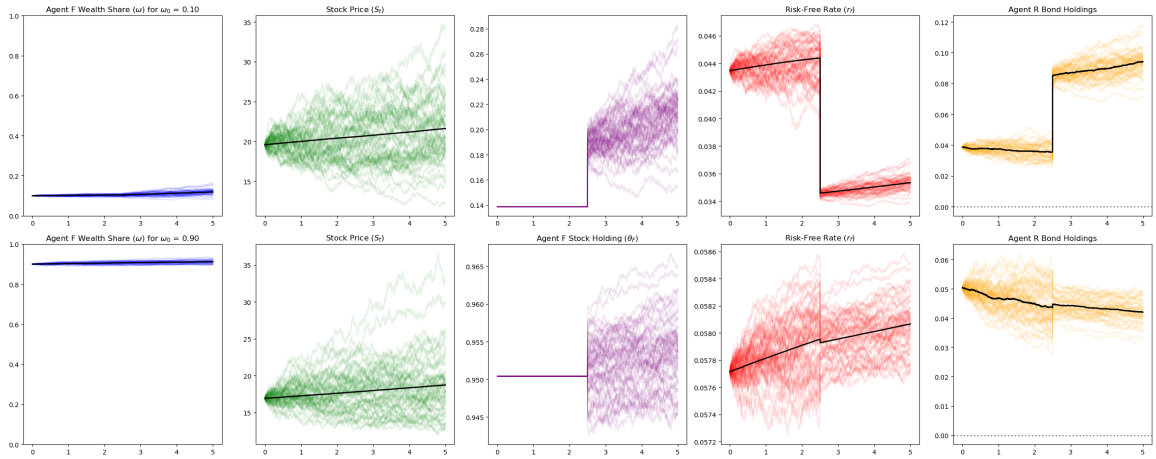


Figure 4: Simulated Infrequent Economy by Initial Wealth Distribution

Table 1: Simulation Outcomes by Initial Wealth Distribution

$\omega_{F,0}$	$\overline{\omega}_F$	$\overline{\omega}_{F,T}$	$\overline{S}$	$\overline{r}_f^{\text{pre}}$	$\overline{r}_f^{\text{post}}$	Δr_f
0.05	0.0536	0.0605	20.821	0.0438	0.0324	-0.0114
0.10	0.1054	0.1170	20.549	0.0459	0.0345	-0.0114
0.15	0.1584	0.1740	20.409	0.0460	0.0366	-0.0095
0.20	0.2096	0.2273	20.150	0.0471	0.0384	-0.0087
0.25	0.2608	0.2801	19.923	0.0479	0.0402	-0.0078
0.30	0.3134	0.3345	19.772	0.0481	0.0419	-0.0063
0.35	0.3644	0.3859	19.558	0.0488	0.0435	-0.0054
0.40	0.4140	0.4352	19.361	0.0499	0.0450	-0.0049
0.45	0.4650	0.4862	19.188	0.0505	0.0465	-0.0040
0.50	0.5147	0.5350	19.015	0.0514	0.0480	-0.0035
0.55	0.5656	0.5851	18.852	0.0519	0.0494	-0.0026
0.60	0.6141	0.6322	18.686	0.0530	0.0507	-0.0023
0.65	0.6637	0.6800	18.520	0.0537	0.0520	-0.0017
0.70	0.7135	0.7285	18.390	0.0544	0.0533	-0.0011
0.75	0.7620	0.7748	18.230	0.0553	0.0545	-0.0008
0.80	0.8104	0.8210	18.064	0.0561	0.0557	-0.0005
0.85	0.8578	0.8660	17.947	0.0572	0.0568	-0.0004
0.90	0.9052	0.9106	17.800	0.0582	0.0579	-0.0003
0.95	0.9525	0.9552	17.630	0.0591	0.0590	-0.0002

When Agent F starts with a small fraction of aggregate wealth, the more risk-averse Agent R effectively controls the asset market. Because Agent R knows they will be locked out of trading for a prolonged period, they face an asymmetric opportunity cost. If they were to hold only their smaller frictionless optimal share at time 0, any subsequent positive economic shocks would increase their wealth, diluting their effective equity exposure, and causing them to miss out on the equity risk premium over the lock-up duration. To self-insure against this inability to dynamically buy more

equity later, Agent R compensates at time 0, optimally choosing to hold a share of the Lucas tree that is larger than what they would otherwise hold in a frictionless market.

This endogenous choice constrains Agent F, forcing the liquid market maker to hold a smaller supply. Consequently, Agent F is starved of equity and forced into an under-levered position relative to their frictionless optimum allocation.

This creates a macroeconomic misallocation of risk: the more risk-tolerant agent F is forced to hold safe bonds, while the more risk-averse agents R left with equity. Because Agent F desperately wants to borrow cash to buy more equity, but is contractually prevented from doing so by the fixed supply, their large demand for leverage drives the risk-free rate high.

When the trading constraint is lifted, agent F immediately purchases their desired equity from Agent R, causing θ_F to jump upward discontinuously. As risk is efficiently transferred back to the more risk-tolerant market maker, Agent F's demand for borrowing decreases, causing the risk-free rate to drop discontinuously to its frictionless equilibrium. As the wealth share of the agent F increases, the jump in risk free rate decreases. Since the liquid agentss are getting richer, the more risk-tolerant agent F holds less bonds, reducing the total amount of the bond transfer when the trading constraint is lifted, and therefore the magnitude of the jump.

Conversely, consider the case where Agent F controls the vast majority of the economy's wealth (bottom row of Figure 4). Because Agent F is more risk-tolerant ($\gamma_F < \gamma_R$) and possesses higher wealth, they are the natural holder of the Lucas tree. Their frictionless, unconstrained optimal portfolio naturally absorbs almost the entire supply of the asset.

In this regime, restricting Agent R's trading capacity is practically irrelevant. Agent R's optimally chosen lock-up share $\bar{\theta}_R^*$ that is very similar to what they would have chosen to hold frictionlessly. Because the constraint barely binds, the restricted economy behaves almost identically to the complete-market baseline. When the trading restriction expires, the transition is smooth, and asset prices exhibit low to no discontinuous jump risk.

The shrinking magnitude of the price jump at $t = T_{vest}$ across the rows of Figure 4 provides a visual representation of the shadow cost of the lock-up constraint. The constraint binds asymmetrically: it is much worse when the market maker is starved of capital, but much less damaging when the market maker is well capitalized. Consequently, if we consider agent F as the market maker or

the intermediary, the ex-ante welfare cost of infrequent trading is not a static penalty, but rather a dynamic cost that scales inversely with the wealth share of the liquid agents. We can think of instances where the liquid agents are starved of capital as liquidity squeezes.

6.6 The Effect of Restriction Time Horizon

While the initial capitalization of each agent determines the magnitude of the pricing correction, we want to investigate the effect of the duration of the friction. By simulating the regime-switching economy across varying trading restriction horizons, we observe an interesting numerical result: the absolute magnitude of the price correction is insensitive to the length of the restricted period.

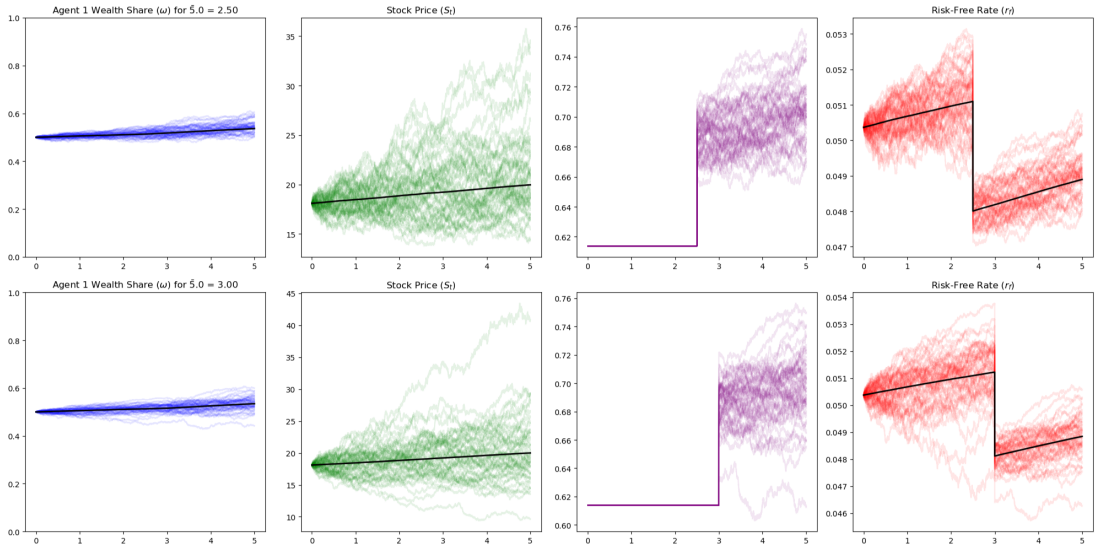


Figure 5: Simulated Infrequent Economy by Restriction Period

The result is analogous to what Merton discovered in the continuous time infinite horizon, that the optimal portfolio choice is time invariant. This constancy is driven by the temporal properties of the agents' optimal ex-ante contract choice. Because the fundamental cash flow dynamics follow a Geometric Brownian Motion with independent and identically distributed shocks, Agent R's optimal "buy-and-hold" portfolio ($\bar{\theta}_R^*$) is practically time-invariant. Whether Agent R faces a 6-month or a 4.5-year trading restriction, their optimal portfolio strategy includes locking in a virtually identical share of the asset.

Because the asset allocation remains static across horizons, the leverage forced upon Agent F and

therefore the general equilibrium pricing required to clear the market is near identical. Consequently, the primary channel through which lock-up duration impacts the economy is not by deepening the severity of the price crash, but by prolonging the duration of the risk misallocation. The ultimate welfare cost of the friction, therefore, scales with the time duration of this pricing distortion, even though the asset price paths look identical.

The expected stock price path is also very similar across different lock-up horizon. This is a direct reflection of the result from Gârleanu 2009. That equilibrium price path is independent from trading restrictions if we let the constrained agent pick how much they want to hold during the illiquid period.

7 Welfare Analysis

7.1 Fictitious Agent

To see the welfare effect of the trading frequency, I introduce a fictitious agent. This agent has the exact same preferences and starting wealth as Agent R (γ_R). However, this fictitious agent is a single individual with measure zero, their individual choices do not affect the equilibrium prices or quantities. They live in the same economy, they observe the distorted prices (r_f, μ_S, σ_S) caused by the aggregate trading restrictions, but they are allowed to trade freely. By calculating the fictitious agent's optimal portfolio (θ_{dev}) and value function (h_{dev}), we can compare them directly to the locked Agent 2 to measure the instantaneous effects of the friction.

The fictitious agent's problem is as follows:

7.2 Fictitious Agent Setup

Let the aggregate state of the economy be defined by Agent F's wealth share, ω_t. During the restricted trading phase, the actual Agent R is restricted to a fixed share of the asset, $\bar{\theta}_R$. This structural friction, combined with the market clearing conditions, determines the equilibrium paths of the risk-free rate $r_f(\omega, t)$, the asset return $\mu_S(\omega, t)$, the asset volatility $\sigma_S(\omega, t)$, and the state dynamics $\mu_\omega(\omega, t)$ and $\sigma_\omega(\omega, t)$.To quantify the welfare cost of this illiquidity, we introduce a fictitious agent (denoted by the subscript $fict$). The fictitious agent satisfy the following assumptions:

Assumption 1 (Measure Zero): The fictitious agent has measure zero. They take the general equilib-

rium prices (r_f, μ_S, σ_S) and state dynamics $(\mu_\omega, \sigma_\omega)$ generated by the economy as strictly exogenous. Their trading behavior does not impact market clearing or prices.

Assumption 2 (Identical Preferences): The fictitious agent possesses the exact same CRRA preferences as Agent R, with risk aversion γ_R and time discount rate ρ.

7.2.1 Fictitious Agent Optimization

The fictitious agent chooses an unconstrained consumption path $c_{fict,t}$ and portfolio leverage $\pi_{fict,t}$ to maximize their expected lifetime utility:

$$V_{fict}(W_{fict}, \omega, t) = \max_{c_{fict}, \pi_{fict}} \mathbb{E}_t \left[\int_t^T e^{-\rho(s-t)} \frac{c_{fict,s}^{1-\gamma_R}}{1-\gamma_R} ds + e^{-\rho(T-t)} U(W_{fict,T}) \right]$$

This is subject to the same standard budget constraint for their wealth W_{fict}:

$$dW_{fict} = W_{fict} \left[r_f + \pi_{fict}(\mu_S - r_f) - \frac{c_{fict}}{W_{fict}} \right] dt + W_{fict} \pi_{fict} \sigma_S dZ_t$$

Because the agent's wealth W_{fict} depends on the state ω, the HJB equation is:

$$\rho V_{fict} - \frac{\partial V_{fict}}{\partial t} = \max_{c_{fict}, \pi_{fict}} \left\{ \frac{c_{fict}^{1-\gamma_R}}{1-\gamma_R} + W_{fict} \left(r_f + \pi_{fict}(\mu_S - r_f) - \frac{c_{fict}}{W_{fict}} \right) \frac{\partial V_{fict}}{\partial W_{fict}} \right.$$

$$\left. + \frac{1}{2} W_{fict}^2 \pi_{fict}^2 \sigma_S^2 \frac{\partial^2 V_{fict}}{\partial W_{fict}^2} + \mu_\omega \frac{\partial V_{fict}}{\partial \omega} + \frac{1}{2} \sigma_\omega^2 \frac{\partial^2 V_{fict}}{\partial \omega^2} + W_{fict} \pi_{fict} \sigma_S \sigma_\omega \frac{\partial^2 V_{fict}}{\partial W_{fict} \partial \omega} \right\}$$

7.3 Decomposing the Cost of Illiquidity: Optionality vs. Price Distortion

Because the zero-measure fictitious agent optimizes their portfolio continuously while the actual restricted agent R is locked into $\pi_R(\omega_t) = \bar{\theta}_R^* / (1 - \omega_t)$, the fictitious agent achieves a weakly higher utility growth factor across the entire state space, $G_{\text{fict}}(\omega, t) \geq G_R(\omega, t)$. Consequently, for $\gamma_R > 1$, their scaled value functions satisfy $0 < h_R^{\text{Baseline}}(\omega_0, 0) \leq h_{\text{fict}}(\omega_0, 0) \leq h_R^{\text{Restricted}}(\omega_0, 0)$.

Comparing the value functions across the three environments, Frictionless Baseline, Restricted, and Fictitious, yields a clean decomposition of the restricted investor's ex-ante certainty equivalent

welfare loss. As derived in Section 7.4, total certainty equivalent welfare loss $\Delta_{\text{total}}(\omega_0)$ is given by:

$$\Delta_{\text{total}}(\omega_0) = 1 - \left(\frac{h_R^{\text{Restricted}}(\omega_0, 0)}{h_R^{\text{Baseline}}(\omega_0, 0)} \right)^{\frac{1}{1-\gamma_R}}. \tag{31}$$

We define the two economic channels as follows:

1. **Loss of Optionality / Partial Equilibrium (PE) Cost (Loss_{PE}):**

$$\text{Loss}_{\text{PE}}(\omega_0) = 1 - \left(\frac{h_R^{\text{Restricted}}(\omega_0, 0)}{h_{\text{fict}}(\omega_0, 0)} \right)^{\frac{1}{1-\gamma_R}}. \tag{32}$$

   This measures the direct individual cost of being unable to dynamically rebalance. Holding the restricted economy's distorted price system fixed, $\text{Loss}_{\text{PE}}(\omega_0)$ is the fraction of initial wealth a single locked-up investor would surrender to regain the personal ability to trade continuously.

2. **General Equilibrium (GE) Price-Distortion Cost (Loss_{GE}):**

$$\text{Loss}_{\text{GE}}(\omega_0) = 1 - \left(\frac{h_{\text{fict}}(\omega_0, 0)}{h_R^{\text{Baseline}}(\omega_0, 0)} \right)^{\frac{1}{1-\gamma_R}}. \tag{33}$$

   This measures the indirect general equilibrium spillover cost caused by collective illiquidity. Even if an individual investor with preferences γ_R is completely free to trade continuously, $\text{Loss}_{\text{GE}}(\omega_0)$ quantifies the fraction of wealth destroyed purely because aggregate lock-ups distort the equilibrium risk-free rate and stock return processes relative to the frictionless baseline.

Note that this decomposition is **exact in multiplicative wealth-retention terms**:

$$1-\Delta_{\text{total}}(\omega_0) = \left(\frac{h_R^{\text{Restricted}}(\omega_0, 0)}{h_{\text{fict}}(\omega_0, 0)} \right)^{\frac{1}{1-\gamma_R}} \left(\frac{h_{\text{fict}}(\omega_0, 0)}{h_R^{\text{Baseline}}(\omega_0, 0)} \right)^{\frac{1}{1-\gamma_R}} = \big(1-\text{Loss}_{\text{PE}}(\omega_0)\big)\big(1-\text{Loss}_{\text{GE}}(\omega_0)\big), \tag{34}$$

which implies

$$\Delta_{\text{total}}(\omega_0) = \text{Loss}_{\text{PE}}(\omega_0) + \text{Loss}_{\text{GE}}(\omega_0) - \text{Loss}_{\text{PE}}(\omega_0)\text{Loss}_{\text{GE}}(\omega_0) \approx \text{Loss}_{\text{PE}}(\omega_0) + \text{Loss}_{\text{GE}}(\omega_0), \tag{35}$$

where the cross-product interaction term $\text{Loss}_{\text{PE}}\text{Loss}_{\text{GE}}$ is second-order (less than 1 basis point across the state space).

7.4 Certainty Equivalent Wealth

What exact percentage of the unconstrained agent's initial wealth would we have to remove to make them exactly as unhappy as the constrained agent? Let W be total wealth. Because the agents have Constant Relative Risk Aversion (CRRA) utility, their total expected lifetime value function V factors into a wealth component and a scaled state-variable component as shown earlier h:

$$V(W, \omega, t) = \frac{W^{1-\gamma}}{1-\gamma} h(\omega, t)$$

We want to find the wealth penalty fraction (let's call it $Loss$) that equates the unconstrained agent's value to the locked agent's value at time $t = 0$:

$$V^{locked}(W, \omega_0, 0) = V^{free}((1 - Loss) \cdot W, \omega_0, 0)$$

Substitute the CRRA value function formula into the equality above using Agent R's risk aversion γ_R:

$$\frac{W^{1-\gamma_R}}{1-\gamma_R} h^{locked}(\omega_0, 0) = \frac{((1 - Loss)W)^{1-\gamma_R}}{1-\gamma_R} h^{free}(\omega_0, 0)$$

The wealth term W and the denominator $(1 - \gamma_R)$ cancel out on both sides, leaving:

$$h^{locked}(\omega_0, 0) = (1 - Loss)^{1-\gamma_R} \cdot h^{free}(\omega_0, 0)$$

Divide by the unconstrained h:

$$\frac{h^{locked}(\omega_0, 0)}{h^{free}(\omega_0, 0)} = (1 - Loss)^{1-\gamma_R}$$

To isolate the $Loss$, raise both sides to the power of $\frac{1}{1-\gamma_R}$:

$$1 - Loss = \left(\frac{h^{locked}(\omega_0, 0)}{h^{free}(\omega_0, 0)} \right)^{\frac{1}{1-\gamma_R}}$$

Finally, rearrange to solve for the exact percentage of wealth lost:

$$Loss = 1 - \left(\frac{h^{locked}(\omega_0, 0)}{h^{free}(\omega_0, 0)} \right)^{\frac{1}{1-\gamma_R}}$$

7.5 Welfare Cost by Risk Aversion

To quantify the economic significance of the infrequent trading friction, we calculate the ex-ante welfare loss as a certainty equivalent wealth. This metric captures the fraction of initial wealth the restricted agent would give up at $t = 0$ to be able to trade continuously instead of being locked away from trading for a period of time in the same economy. Figure 6 plots this welfare loss across the parameter space of relative risk aversions (γ_F, γ_R).

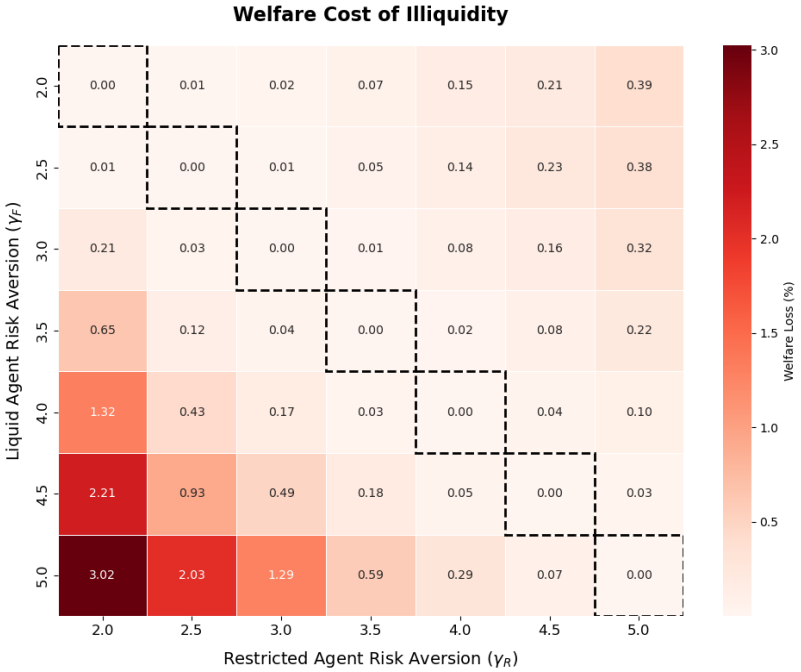


Figure 6: Welfare Cost of Illiquidity

If we look at the highlighted diagonal entries of the heatmap, when $\gamma_F = \gamma_R$, the trading restriction is almost costless because the intertemporal motive for trade is zero. This is a numerical verification from the theoretical result of Gorman aggregation that we will prove in the next section. Without preference heterogeneity, the restricted agent's optimal ex-ante contract perfectly matches the unconstrained buy-and-hold strategy, rendering the lock-up constraint entirely non-binding.

Once we go off diagonal though, the trading restriction is no longer costless. As we move further and further away from the diagonal, that is if the agents are becoming more and more different in their risk aversion, the welfare cost also grow larger. The more different the agents are, the larger the welfare loss.

However, the effect is not symmetrical. The most significant welfare losses (exceeding 3.00%) occur in the lower-left quadrant, where the liquid market makers are more risk-averse ($\gamma_F = 5.0$) and the restricted agents are more risk-tolerant ($\gamma_R = 2.0$). Because the restricted agent is more risk-tolerant, they optimally lock in a large equity position. While this initially leaves a small Lucas tree share for the market maker, adverse macroeconomic shocks ($\omega \to 0$) can still force the liquid trader into highly levered positions. Because the liquid market maker is highly risk-averse, forcing them to absorb any residual leverage requires a very large equity premium. This pricing squeeze imposes much greater welfare penalties on the locked-up agent.

Conversely, in the upper-right quadrant ($\gamma_F = 2.0, \gamma_R = 5.0$), the welfare loss is an order of magnitude smaller (roughly 0.39%). Here, the liquid market maker is highly risk-tolerant. A risk-averse restricted agent will conservatively lock in a small equity share, forcing the liquid traders to take on the vast majority of the asset. However, because the liquid traders possess a higher risk preference, they are more willing to absorb this excess supply without demanding a much higher risk premium. The pricing squeeze is largely eliminated by the market maker's willingness to hold risk. The restricted agent suffers a minor loss of optionality, but largely avoids the loss of welfare coming from market mispricing.

This asymmetry reveals that illiquidity frictions are systemically significant primarily when the remaining liquid market makers are highly risk-averse. The health of the market depends heavily on the risk appetite and balance sheet capacity of the unconstrained liquid traders.

7.6 Gorman Aggregation and the Irrelevance of Trading Constraints

To formally demonstrate why the welfare cost of illiquidity disappears when agents have identical risk preferences ($\gamma_F = \gamma_R = \gamma$), I compare the complete-market equilibrium to the restricted trading equilibrium under exact aggregation.

Assume a complete market where both agents have identical CRRA preferences ($\gamma_F = \gamma_R = \gamma$). Because the investment opportunity set is identical for both agents, their value functions take the same form, and their optimal consumption-to-wealth ratios are identical: $\tilde{c}_F = \tilde{c}_R = \tilde{c}$.

From the unconstrained Hamilton-Jacobi-Bellman (HJB) equations, the optimal portfolio choice for both agents is given by the standard Merton rule. Because the wealth distribution ω_t ceases to be a

priced state variable under Gorman aggregation, both agents have zero hedging demand ($\frac{\partial V_i}{\partial \omega} = 0$):

$$\pi_F = \pi_R = \frac{\mu_S - r_f}{\gamma \sigma_S^2}$$

The asset market clearing condition requires that total wealth invested in the risky asset equals the total value of the asset:

$$\omega_t \pi_F + (1 - \omega_t)\pi_R = 1$$

Substituting the identical optimal portfolios into the clearing condition yields:

$$\omega_t \left(\frac{\mu_S - r_f}{\gamma \sigma_S^2} \right) + (1 - \omega_t) \left(\frac{\mu_S - r_f}{\gamma \sigma_S^2} \right) = 1 \implies \pi_F = \pi_R = 1$$

Because both agents continuously choose to hold a portfolio weight of exactly 1.0 (the market portfolio), their shares of the Lucas tree are perfectly proportional to their wealth:

$$\theta_F = \omega_t \quad \text{and} \quad \theta_R = 1 - \omega_t$$

By applying Itô's Lemma, the evolution of Agent 1's wealth share ω_t is determined by the relative difference in portfolio choices and consumption. The endogenous volatility of the wealth distribution is:

$$\sigma_\omega = \omega_t(1 - \omega_t)(\pi_F - \pi_R)\sigma_S$$

Because $\pi_F = \pi_R = 1$, the diffusion collapses to zero ($\sigma_\omega = 0$). Furthermore, because their consumption-to-wealth ratios are identical ($\tilde{c}_F = \tilde{c}_R$), the drift of the wealth share also collapses to zero ($\mu_\omega = 0$).

Therefore, in the unconstrained homogeneous economy, the wealth distribution is perfectly static. For all $t > 0$:

$$\omega_t = \omega_0 \quad \text{and} \quad d\omega_t = 0$$

Now, consider the restricted economy where Agent R is locked into a fixed share of the asset at $t = 0$, matching their initial wealth:

$$\bar{\theta}_R = 1 - \omega_0$$

Under this restriction, Agent R cannot trade. Agent F, as the residual market maker, must hold the remaining supply of the asset: $\theta_F = \omega_0$.

Agent F's constrained leverage is therefore determined by the market clearing condition:

$$\pi_F = \frac{\theta_F}{\omega_t}$$

Because the economy begins at ω_0 and the representative agent pricing kernel ensures $\mu_\omega = 0$ and $\sigma_\omega = 0$, the wealth distribution never drifts from its initial state ($\omega_t = \omega_0$). Substituting this into Agent 1's forced leverage:

$$\pi_F = \frac{\omega_0}{\omega_0} = 1$$

In the restricted economy, Agent F is the sole marginal pricer of the asset. Agent F dictates the market price of risk via their constrained first order condition:

$$\mu_S^{(1)} - r_f^{(1)} = \gamma \sigma_S^2 \pi_F$$

Because the lock-up physically forces Agent 1 into a leverage of $\pi_F = 1$, the equilibrium expected excess return becomes:

$$\mu_S^{(1)} - r_f^{(1)} = \gamma \sigma_S^2$$

This is exactly identical to the expected excess return in the complete-market baseline.

Therefore, when $\gamma_F = \gamma_R$, the Gorman aggregation suggests that the optimal strategy is buy and hold. The contractual lock-up $\bar{\theta}_R = 1 - \omega_0$ perfectly coincides with the unconstrained optimal policy $\theta_R^* = 1 - \omega_t$. Because Agent 2 would have endogenously chosen to hold the exact amount they are contractually forced to hold, and because they have no desire to dynamically rebalance, the constraint is strictly non-binding. The transition matrix, equilibrium prices, and resulting value functions $V^{(1)}$ and $V^{(2)}$ are mathematically indistinguishable, resulting in an ex-ante welfare cost of exactly zero.

7.7 Welfare Cost by Wealth Distribution and Restriction Horizon

We then plotted the welfare loss by wealth distribution and the trading restriction duration to study the effects of each.

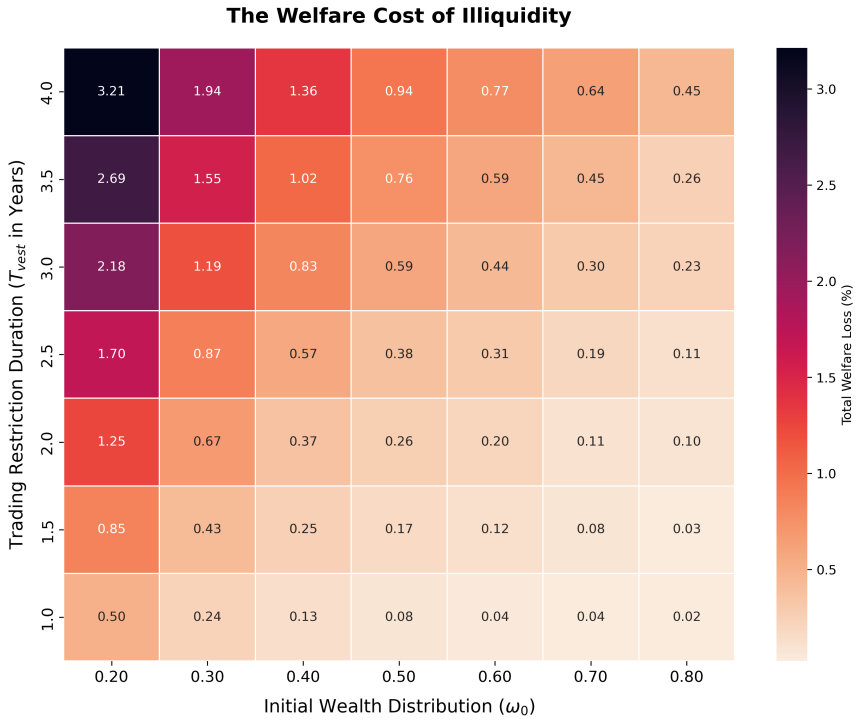


Figure 7: Welfare loss by wealth distribution and the trading restriction duration

On the right side of the heatmap where we have high ω_0, regardless of how long the lock-up lasts, the welfare loss is near zero. When the liquid agents controls the vast majority of the economy's wealth, the restricted agent's optimal lock-up mostly matches their frictionless target. The constraint barely binds, making the friction almost harmless.

In the top-left corner where we have low ω_0 and long restriction period, Agent R controls the wealth and hoards the asset to capture the premium over a long horizon, starving Agent F of equity. The mismatch causes more significant price distortions.

Initial Wealth (ω_0)	Stock Volatility (σ_S %)	Risk-Free Rate (r_f %)	Welfare Loss (%)
0.20	9.48%	2.50%	3.21%
0.50	8.59%	4.37%	0.94%
0.80	9.21%	5.45%	0.45%

Table 2: Unconditional Moments ($T_{restrict} = 4.0$)

7.7.1 Quantifying the Welfare Cost in Context

Figure 7 illustrates the central quantitative takeaway of the paper: *a trading lock-up is nearly free when the liquid sector is well-capitalized and expensive when it is not.* Holding the lock-up duration fixed at four years ($T_{\text{restrict}} = 4.0$), when the liquid trader commands 80% of initial aggregate wealth ($\omega_0 = 0.80$), the ex-ante certainty-equivalent welfare cost to the restricted investor is less than 0.5% (and just 0.11% for a 2.5-year lock-up). Because a well-capitalized, risk-tolerant market maker naturally absorbs almost the entire equity supply, the restricted investor's optimal locked share $\bar{\theta}_R^*$ closely tracks their frictionless target, leaving equilibrium prices virtually undistorted.

Conversely, when the liquid sector is poorly capitalized, holding only 20% of initial aggregate wealth ($\omega_0 = 0.20$), the welfare cost of the same four-year lock-up surges more than sixfold to over 3.2% (and 1.70% for a 2.5-year lock-up). To place this magnitude in context, recall that in aggregate macro finance, certainty equivalent welfare losses measure the permanent fraction of initial wealth an investor would surrender at $t = 0$ to avoid the friction. In his seminal benchmark, Lucas (1987) demonstrated that eliminating all U.S. business cycle consumption fluctuations is worth only 0.05% to 0.10% of lifetime consumption under standard risk aversion. When liquidity providers are capital-scarce, imposing a single multi-year trading lock-up destroys over thirty times more certainty-equivalent welfare for the restricted investor than the classic Lucas cost of business cycles. For an institutional endowment or pension fund, surrendering over 3.2% of total capitalized fund value represents a first order economic penalty.

Crucially, this 3.21% loss is incurred after the restricted investor has optimally self-insured at $t = 0$ by choosing $\bar{\theta}_R^*$ to anticipate the lock-up and its equilibrium price path. If the restricted investor were instead locked into a naive or exogenous allocation without date-zero optimization, the resulting welfare loss would be substantially larger. Thus, our framework isolates the irreducible general equilibrium shadow cost of horizon-bound illiquidity.

7.8 Welfare Cost by GE vs PE

Lastly we plot the welfare cost due to optionality alone versus the cost due to market mispricing.

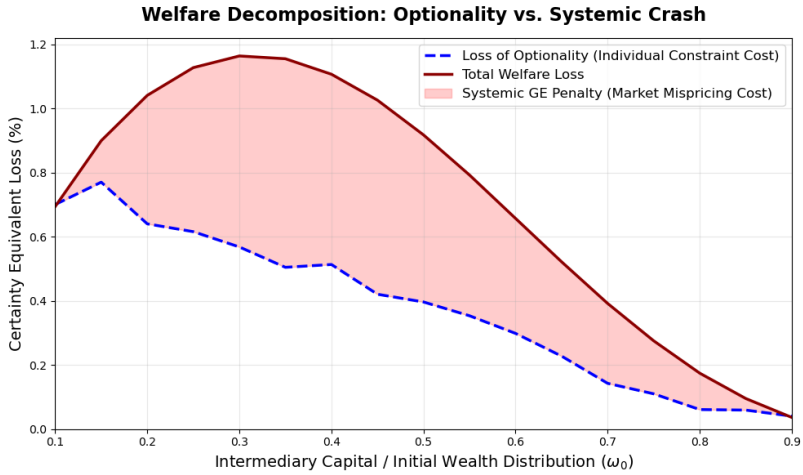


Figure 8: Welfare loss from PE vs GE

The first thing to notice is that the effect is highly asymmetrical. At regions closer to the boundaries, since one agent owns most of the economy, we revert back to the buy and hold optimum and the price distortion is minimal since they would have chosen a buy and hold strategy anyway. This is another numerical result of the Gorman aggregation shown in the previous chapter. The effect is also larger for both PE and GE on the left hand side of the graph, which confirms the result above that a poor liquid agent is worse than a poor restricted agent. This graph also visually confirms the fact that the GE effect is always larger than the PE effect across the interior.

7.9 Summary of Numerical Implementation

Because the HJB equations are non-linear, and asset prices depend on the spatial derivatives of the value functions, while the value functions simultaneously depend on asset prices, we must solve the system iteratively. The numerical solution algorithm proceeds in three primary steps:

- Step 1: Initialization

  I establish the state space grid $\omega \in (0, 1)$ with J discrete nodes. At time-to-maturity $\tau = 0$ (calendar time $t = T$), I initialize the numerical value functions using the closed-form terminal condition: $h_i^0(\omega) = K_i^{-\gamma_i}$ for $i \in \{F, R, Fictitious\}$. Note here that we are solving for the

hypothetical zero measure agent simultaneously alongside the rest of the population.

- Step 2: Implicit Time-Stepping

  For each time step n from $\tau = 0$ to $\tau = T$ (stepping backward in calendar time), we perform the following sequence:

  Step 2a Prices and Policies: Using the known value functions h_i^n from the current step, we compute the spatial derivatives $\partial h/\partial \omega$ using standard finite differences. We then solve the market clearing conditions to find the endogenous general equilibrium asset prices $(r_f^n, \mu_S^n, \sigma_S^n)$ and the corresponding optimal consumption and portfolio policies $(\tilde{c}_i^n, \pi_i^n)$.

  Step 2b Matrix Construction: Using these newly computed prices and policies, we calculate the Utility Growth Factor G_i^n and the Effective Drift μ_ω^{eff}. We determine the positive and negative drift components (μ^+, μ^-) to construct the sparse, tridiagonal transition matrix $\mathbf{A}^n$ via the upwind scheme.

  Step 2c Matrix Inversion: I assemble the M-Matrix: $\mathbf{M}^n = \mathbf{I}\big(1 + \Delta\tau(\rho - \mathbf{G}_i^n)\big) - \Delta\tau\mathbf{A}^n$ and solve the linear system $\mathbf{M}^n\mathbf{h}_i^{n+1} = \mathbf{b}^n$ using a standard sparse system solver to obtain the updated value functions for the next time step, h_i^{n+1}.

- Step 3: Ex-Ante Optimization

  Steps 1 and 2 yield the equilibrium price paths and the initial value function $V_R(\omega_0, 0)$ for a given lock-up constraint $\bar{\theta}_R$. To find the full general equilibrium, I wrap the entire PDE solver in a global optimization routine. We sweep across the admissible space of asset allocations $\bar{\theta}_R \in (0, 1)$, solving the full PDE for each candidate, and select the unique $\bar{\theta}_R^*$ that maximizes Agent R's ex-ante expected utility at $t = 0$.

8 Conclusion

The assumption of continuous, frictionless trading has long served as the mathematical foundation of asset pricing. Yet, the modern financial landscape is increasingly characterized by capital that is contractually frozen or passive, whether through household behavior, private equity lock-ups, passive funds, hedge fund gates, or institutional vesting requirements. By developing a finite-horizon, continuous-time heterogeneous agent model, this paper provides a tractable framework to analyze the

equilibrium dynamics of these explicit, time-bound trading restrictions, overcoming the numerical instabilities that have historically limited infinite-horizon approaches.

Our analysis demonstrates that a trading lock-up is nearly free when the liquid sector is well-capitalized and expensive when it is not, and that most of this cost stems from the general equilibrium price distortion the constraint creates rather than from the inability to trade. When restricted investors face a multi-year lock-up, they self insure against missing out on future equity premia by holding onto extra equity at inception. This precautionary hoarding starves the liquid trader of equity, forcing the unconstrained market maker into an under levered position, elevating the equilibrium risk-free rate, and weakening the endogenous price smoothing of the frictionless market. At the moment the lock-up expires, pent-up risk-sharing demand triggers an instantaneous block trade from the restricted investor to the liquid trader and a sharp downward jump in the risk-free rate.

Mapping the ex-ante certainty-equivalent welfare cost to the restricted investor across the state space reveals a striking asymmetry. For a four-year lock-up, when the liquid sector holds 80% of initial aggregate wealth, the friction is nearly costless, generating a welfare loss below 0.5% (0.45%). When the liquid sector holds only 20% of initial wealth, however, the welfare cost surges more than sixfold to over 3.2% (3.21%), dwarfing the classic Lucas (1987) cost of business cycle fluctuations. Furthermore, our multiplicative welfare decomposition via a zero-measure shadow agent proves that when intermediary capital is scarce, the majority of this welfare loss arises from the general equilibrium price distortion imposed on the market rather than from the partial equilibrium loss of rebalancing optionality.

These findings have direct implications for the regulation of shadow banking and the ongoing structural shift toward private markets and passive investment vehicles. As a growing fraction of global wealth becomes locked into rigid structures and more capital becomes passive, the effective capitalization of the remaining liquid market makers becomes increasingly important. A natural avenue for future research is to extend this finite-horizon framework to include multiple risky assets with varying degrees of liquidity, or to allow for endogenous capital injections into the liquid agents. Ultimately, this paper highlights that in an economy with restricted participation, the resilience of asset prices depends not on the average investor, but largely on the capacity of the investors who remain free to trade.

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A Simulated Infrequent Economy Full Panels

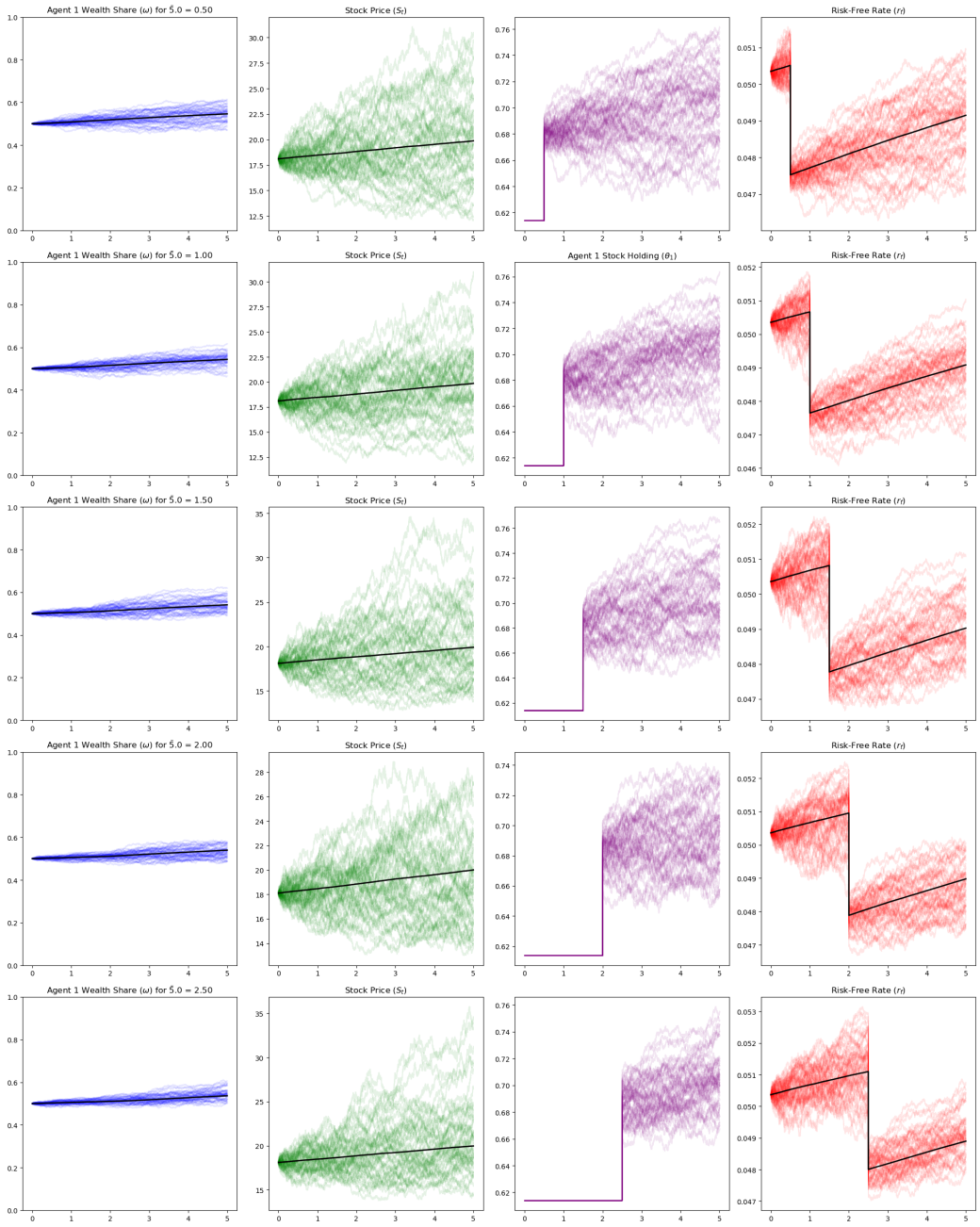


Figure 9: Simulated Infrequent Economy by Restriction Period Full Panel Part 1

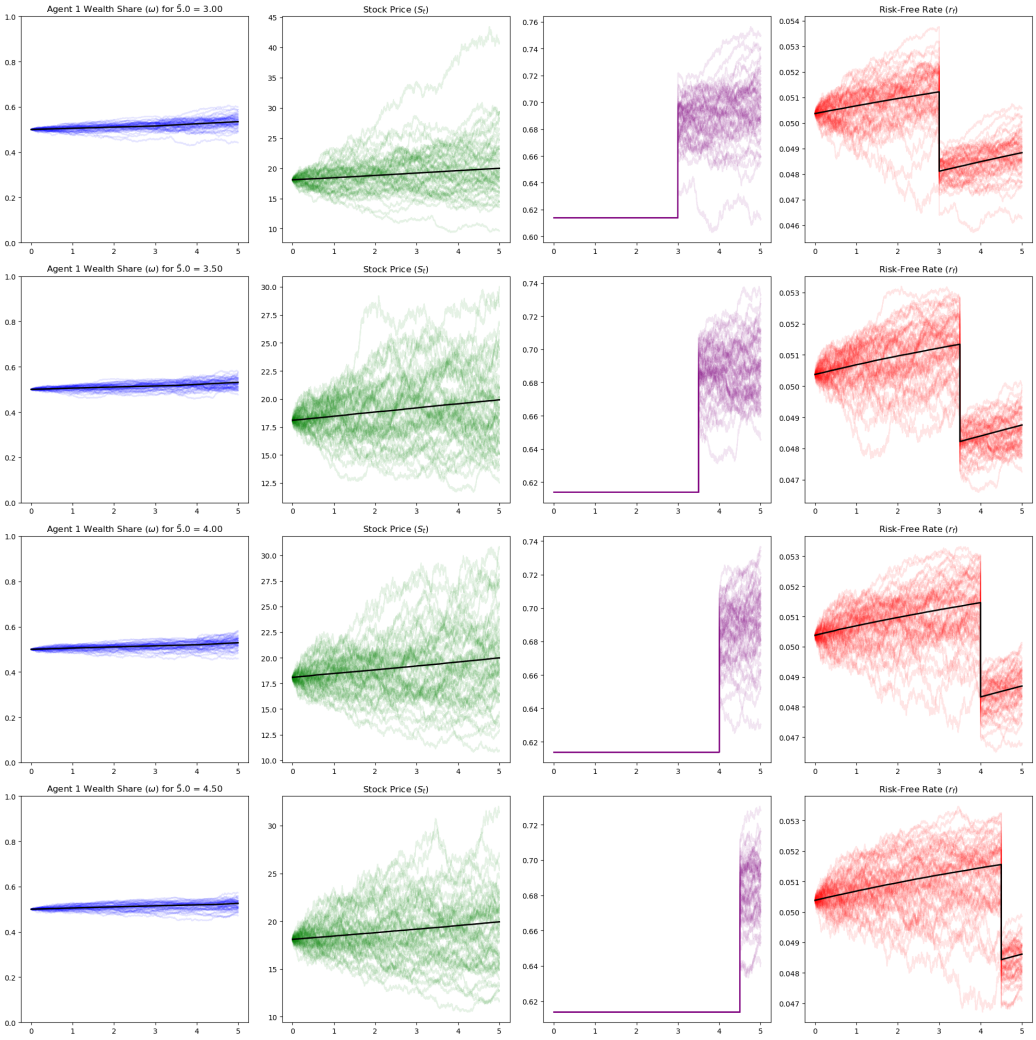


Figure 10: Simulated Infrequent Economy by Restriction Period Full Panel Part 2

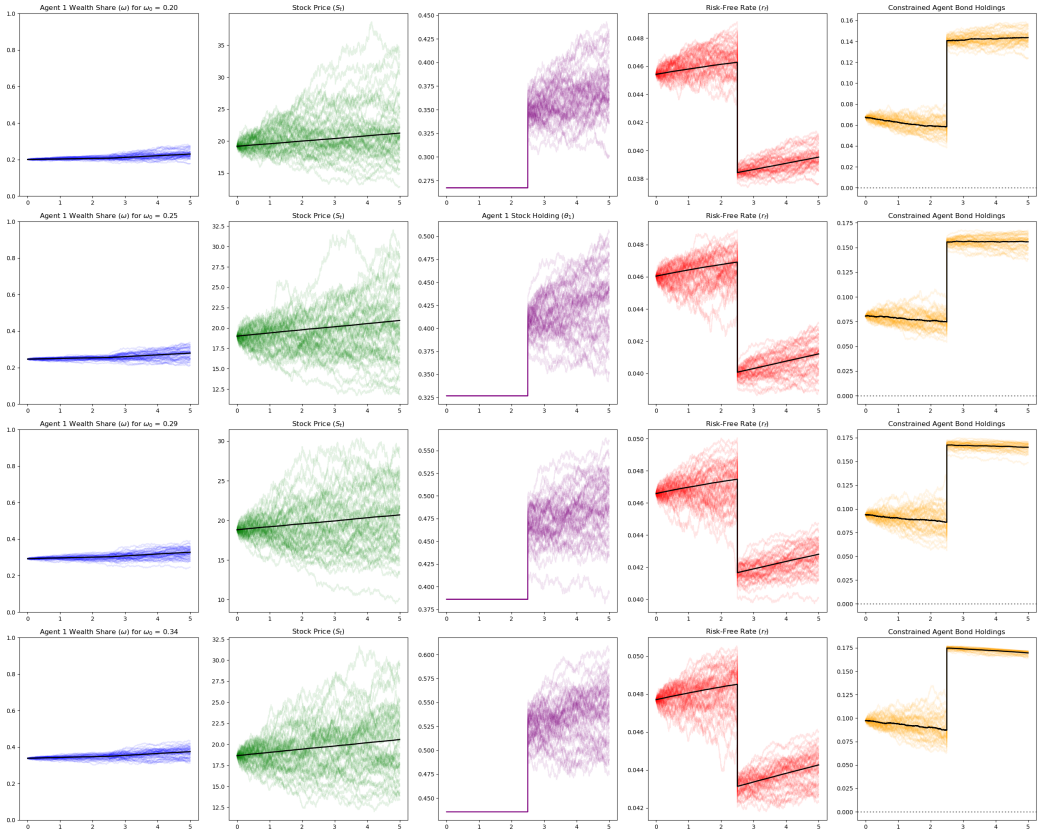


Figure 11: Simulated Infrequent Economy by Initial Wealth Distribution Full Panel Part 1

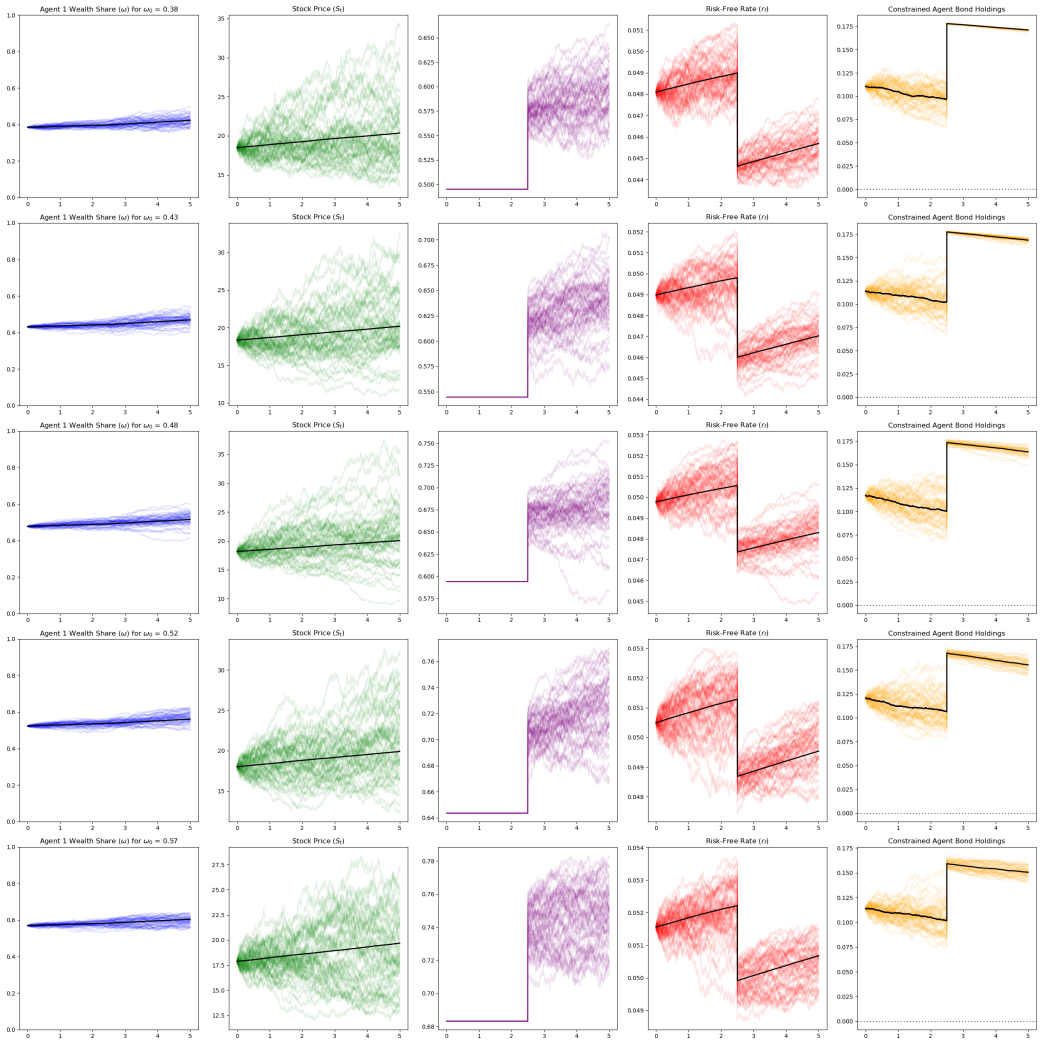


Figure 12: Simulated Infrequent Economy by Initial Wealth Distribution Full Panel Part 2

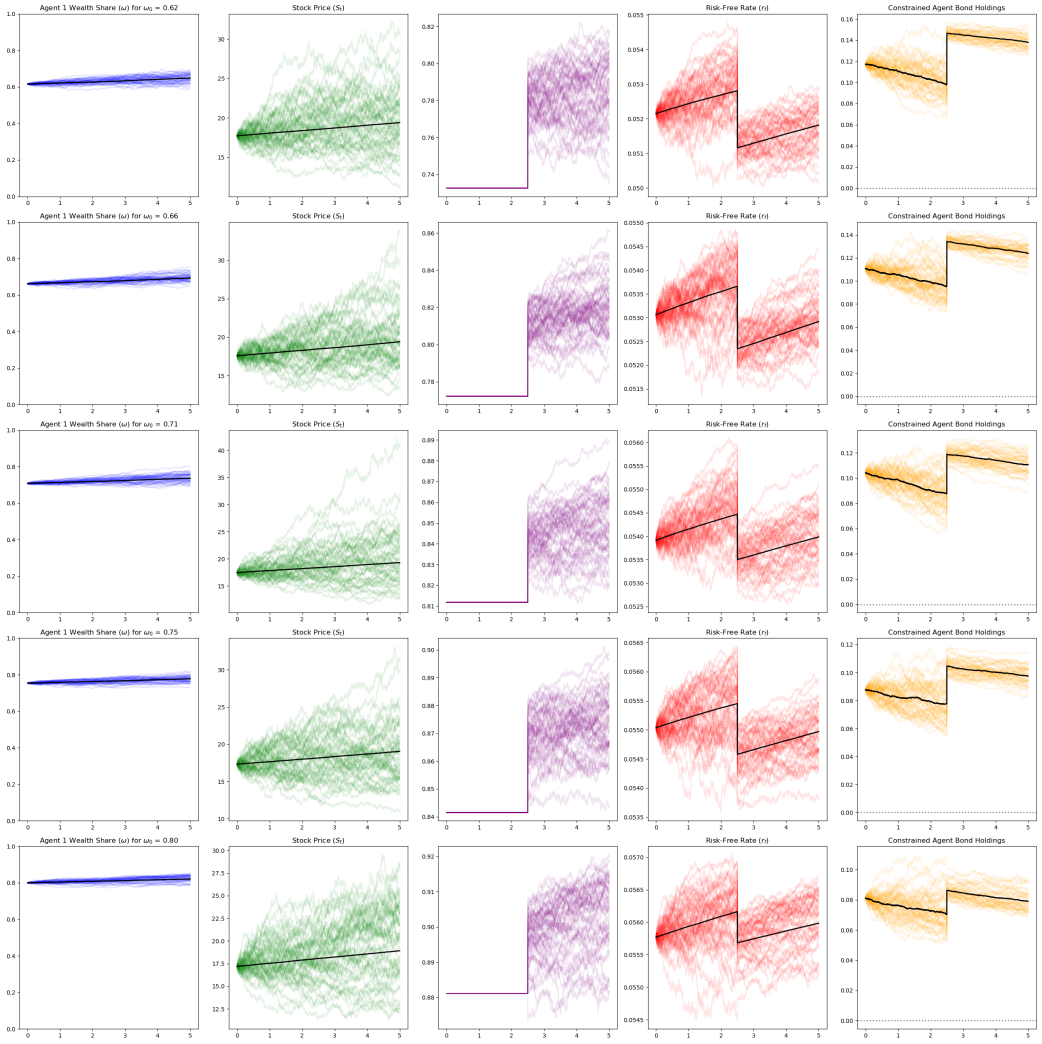


Figure 13: Simulated Infrequent Economy by Initial Wealth Distribution Full Panel Part 3